MILITARY SPECIFICATION

Control, Radio Set
C-10319A/ARC, C-10776A/ARC, and C-11131/ARC

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, and test requirements for the C-10319A/ARC, C-10776A/ARC, and C-11131/ARC Radio Set Controls hereinafter referred to as the controls. The controls covered in the specification are used to control the receiver-transmitters of the ARC-182 family. The principle difference between the three controls is the colors of the panel lighting which are red for the C-10319A/ARC, white for the C-10776A/ARC, and blue-white for C-11131/ARC.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions,) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 53, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

AMSC N/A

FSC 5821

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Federal

QQ-P-416	Plating

Military

MIL-B-5087	Bonding, Electrical, and Lightning Protection, for Aerospace Systems
MIL-W-5088	Wiring; Aerospace Vehicle
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-C-25050	Colors, Aeronautic Lights and Lighting Equipment, General Requirements for
MIL-M-38510	Microcircuits General Specification for
MIL-C-38999	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect, Environmental Resistant, Removable Crimp Contacts; General Specification for
MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities
MIL-P-55110	Printed Wiring Boards

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

FED-STD-595	Federal Standard Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-415	Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-454	Electronic Equipments, General Requirements for
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Aircraft, Electric Power Characteristics
MIL-STD-781	Reliability Tests; Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810	Environmental Test Methods
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-965	Parts Control Program
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1553	Aircraft Internal Time Division Command/Response Multiplex Data Bus
MIL-STD-1562	Lists of Standard Microcircuits
MIL-STD-2074	Failure Classification for Reliability Testing
MIL-STD-2076	Unit Under Test Compatibility with Automatic Test Equipment, General Requirements for
MIL-STD-2084	Maintainability of Avionics and Electronic Systems and Equipment, General Requirements for

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DOCUMENTS

Naval Air Systems Command

EI-669	Avionics Installation Instructions for Radio Set AN/ARC-182
WS-6536	Process Specification Procedures and Requirements for Preparations and Soldering of Electrical Connections

Department of the Navy

NAVMAT P-4855.1	Navy Power Supply Reliability
NAVMAT P-9492	Navy Manufacturing Screening Program

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item description. The C-10319A/ARC, C-10776A/ARC, and C-11131ARC Radio-Set Controls shall be self-contained units suitable for cockpit mounting. The controls shall contain all the circuits and control facilities necessary to control a host receiver-transmitter and optical indicators from a remote location.

3.2 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.3 Parts. In the selection of parts, the prime considerations are fulfillment of performance requirements, logistics supportability and reliability assurance in a cost effective design. The following shall govern in parts selection.

a. Subassemblies shall be designed and fabricated to be repairable. Maximum economic standardization of parts and materials shall be exercised. Parts selection shall be made in the following order of precedence:

(1) Military Standard Parts shall be selected in accordance with MIL-STD-454.

(2) Nonstandard parts previously qualified for other programs.

(3) Nonstandard parts.

b. Nonstandard parts and nonestablished reliability military standard parts shall be screened to the requirements of the most similiar military standard or standard nonestablished reliability part. Integrated circuits and hybrids shall use MIL-STD-883, Method 5004, Class B, as a basis for tailored screens.

3.3.1 Component rescreening. The contractor shall conduct 100 percennt incoming inspection of all active components. Functional and parametric checks of active components will be performed at the components rated temperature extremes and at ambient. Components which are not compliant with their specifications shall not be used in the ARC-182.

3.3.2 Parts derating and application. All parts used shall be applied well within their ratings. The derating shall encompass the appropriate and meaningful application conditions such as voltage, current, power, temperature, mechanical, and duty cycle. Electronic and electromechanical parts shall conform to Table I electronic parts derating for worst case electrical and environmental stress unless formal written approval is received from the procuring activity prior to incorporation into the design. Part level stress analysis shall be used to verify that all parameter stresses are within the derated values at worst case circuit and environmental conditions. The following limitations on parts usage shall apply:

a. Prohibited application

(1) Non-hermetically sealed semiconductors except power transistors
(2) Photo couplers (photo transistors)
(3) Hot carrier (Schottky) rectifiers
(4) Microcircuit sockets
(5) Non-hermetically sealed wet tantalum capacitors

3.3.3 Parts control. The contractor shall establish and maintain a parts control program in accordance with the requirements of MIL-STD-965, Procedure I and as specified herein. The program shall control part standardization and the criteria for the selection, application, and testing of parts to achieve the reliability required. As a minimum, a parts program equal to that specified herein shall be required of subcontractors and suppliers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE I. Electronic parts derating.

Part Type	Parameter	Maximum Allowable Application Stress
Capacitor	Applied Voltage (Ripple Voltage for Tantalum)	50% of rating 1/
Resistors	Power Voltage	50% of rating 70% of rating
Diodes	Power Current Voltage Junction Temperature	50% of rating 75% of rating 60% of rating 2/
Transistors	Power Voltage Junction Temperature	55% of rating 60% of rating 2/
Integrated Circuits		
Digital	Fan-Out, Fan-In Junction Temperature	80% of rating 2/
Linear	Power Junction Temperature	75% of rating 2/
Transformers, Inductive Devices	Parts shall be used at a winding temperature of at least 30 degrees below rated temperature. In addition, the maximum voltage between windings or between a winding and a shield shall be not greater than 60% of rated voltage for the steady state conditions and 90% of rated voltage with transients. General purpose inductors, audio and power transformers shall be applied at no more than 70% of rated current.	

1/ For solid tantalum and all electrolytic capacitors, the maximum applied voltage shall be not greater than 70 percent of the vendor rated value. "Applied voltage" is the sum of the applied peak ripple voltage and the applied DC voltage.

2/ Semiconductor devices shall be mounted and operated in such manner to insure worst-case junction temperature is not greater than 110 degrees C at +71 degrees C ambient temperature. Any deviation requires procuring activity approval. Power semiconductor devices, on a very selective basis, may be mounted and operated with worst case junction temperatures not to be greater than 150 degrees C at +71 degrees C ambient temperatures.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

3.3.4 Non-standard parts approval. The contractor shall obtain procuring activity approval of all non-standard parts used in the equipment. Standard parts are defined in MIL-E-5400. In addition to the MIL-E-5400 requirements for standard parts, microelectronic parts are standard only if:

a. The device is listed in MIL-STD-1562, and

b. Sources (at least two) are listed on the Qualified Parts List (QPL), and

c. The part is procured from one of the qualified sources, and

d. Quality level "B" (or better) of MIL-M-38510 is required.

The approval procedure shall be in accordance with MIL-E-5400 and MIL-STD-965, Procedure I. In addition to the requirements of MIL-E-5400, approval of non-standard parts will be based on:

a. Suitability for the application

b. Conformance to this specification and to MIL-E-5400

c. Interchangeability

d. Screening (burn-in) method

3.3.5 Samples of non-standard parts. Samples of non-standard parts are not required unless specified in the contract. Where non-standard parts samples are specified in the contract, the sample shall be in accordance with MIL-E-5400 (see 6.2.1).

3.3.6 Cabling and connections.

3.3.6.1 Cables and connectors. The controls shall provide for the use of cables and connectors in accordance with MIL-E-5400. The design of connectors shall be such that improper assembly, mating or installation is impossible. Non-RF connectors shall meet the requirements of MIL-C-38999. All external connectors shall be scoop-proof, have cadmium plating finish in accordance with QQ-P-416 (over a suitable underplating) to withstand a 500 hour salt spray test, and provide 360 degrees circumferential grounding techniques for grounding prior to contact engagement.

3.3.6.2 Interconnection cabling. The controls shall be capable of required operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the controls.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

3.3.7 Control panels. The control panels shall conform to the applicable requirements of MIL-C-6781.

3.3.7.1 Lighted panels-console mounted equipment. The plastic panels for all console mounted control panels, indicators, and radio sets shall conform to MIL-P-7788, Class 1-R, 1-W, or 1-BW Type V, with both 5- and 28-volt (ac and dc) lamp systems installed. Separate input power pins shall be provided for the two lighting systems. Color of panel lighting shall be in accordance with MIL-C-25050, as selected from the options by nomenclature assignment or by description on purchase orders listed in Table II.

TABLE II. Panel lighting color options

Nomenclature	Color (Short Name)	Color In Accordance With MIL-C-25050	MIL-P-7788 Class
C-10319A/ARC	IPL Red	Type 1 (f)	1-R
C-10776A/ARC	IPL White	Type 1 (g)	1-W
C-11131/ARC	Blue Filtered White	Type 1 (h)	1-BW

3.3.7.2 Console mounted equipment function marking. Console mounted control panels shall be permanently and clearly marked to indicate the function of the equipment with which they are associated. Console mounted control panels shall be identified by the abbreviation VHF/UHF located on the front panel.

3.3.7.3 Control knobs and functions. The number of controls required to operate the equipment shall be limited to and consistent with performing all the functions required by this specification. Closely associated controls such as control knobs on concentric shafts, shall be easily identified by contrasting colors of black and gray.

3.3.8 Interchangeability. Physical and functional interchangeability shall exist for all units and replaceable assemblies, subassemblies, and parts of the controls in accordance with MIL-E-5400. The controls shall be constructed to preclude the requirement for any adjustment when replacing an item by another item of the same type designation.

3.4 Design and construction. The controls shall conform with all the applicable requirements of MIL-E-5400 for design and construction except where otherwise specified herein. In any case, it is a condition of final acceptance that the controls shall meet all the performance requirements of this specification.

3.4.1 Reliability. When specified in the contract or purchase order, the contractor shall conduct a reliability program using MIL-STD-785 (see 6.2.1, 6.2.2). Task 104 of MIL-STD-785 shall be implemented to establish and maintain a closed-loop failure reporting system. This system will identify failure trends and patterns at all levels of assembly prior to acceptance of the hardware and will also identify failure trends related to field failures

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

throughout the contract period. The contractor shall develop and implement corrective action for significant failure trends. The reliability program shall include provisions for assuring the reliability of the control or any portion of the equipment supplied for use as spare or repair part.

3.4.1.1 Operational stability. The controls shall operate with required performance for at least the specified mean (operating) time between failures without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.4.1.2 Operating life. The controls shall have a total operating life of 20,000 hours with minimal servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.4.1.3 Reliability in MTBF (mean time between failure). The equipment including any built-in-test (BIT) provisions, shall have a specified mean (operating) time between failures, upper test MTBF (θ_0) of 7500 hours (as specified in MIL-STD-781), and a lower test MTBF (θ_1) of 3750 hours when tested and accepted in accordance with 4.4.5.

3.4.2 Maintainability. The maintainability program, built-in test features, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in MIL-STD-2084. The fault detection percentage requirements of MIL-STD-2084 shall be 95 percent minimum of the critical path circuitry.

3.4.2.1 Compatibility with ATE. The control shall be compatible with Automatic Test Equipment (ATE) for WRA testing and fault isolation to a faulty SRA in accordance with MIL-STD-2076.

3.4.2.2 Maintainability design. All electrical or electronic circuits and parts shall be packaged on replaceable and repairable, plug-in modules. Circuit breakers, indicators, and other items which may logically be bolted and soldered, or otherwise more rigidly electrically and mechanically fastened to control housing shall not be considered modules. SRAs (Shop Replacement Assemblies) shall be QRA's (Quick Replaceable Assemblies) as specified in MIL-STD-2076. SRA arrangement shall be such that access to any SRA does not require the removal of adjacent SRAs or parts other than access panels. The use of a special tool or tools for SRA removal is prohibited. Fuses may be used in lieu of circuit breakers.

3.4.2.2.1 Keying. Connectors shall be keyed to prevent the insertion of a connector or subassembly into an improper location within or on the controls. All connectors shall be legibly marked with reference designations.

3.4.2.2.2 Access panels. Access panels shall be retained by fasteners suitable to the procuring activity.

3.4.2.2.3 Service access. Devices requiring removal for inspection and servicing shall be retained by fasteners suitable to the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

3.4.2.2.4 Adjustments. No adjustment and alignment shall be required at the organizational level. Adjustment and alignment if required at a higher maintenance level (intermediate or depot), shall not require the removal of the part to be adjusted or aligned, or of the hardware element to which it is attached.

3.4.2.2.5 Test points. All test points shall be in accordance with MIL-STD-2084 and MIL-STD-415 to accommodate approved test equipment. MIL-STD-2084 takes precedence over MIL-STD-415 requirements. The SRA test points and related connector requirements of MIL-STD-2084 are waived where prohibited by space or circuit performance considerations. Conformal coating may be applied to PC pads used as test points. Injection test points in the Built-In-Test circuits are not required. The SRA test points and related connector requirements philosophy shall be reflected in appropriate maintenance data. Conformal coating may be applied to the PC pads used as test points. Injection test points in the BIT circuits are not required.

3.4.2.3 Organizational level maintainability. The controls shall provide the operator or organizational maintenance technician, while airborne or on the ground with performance or readiness test and fault (GO/NO-GO) indication.

3.4.2.3.1 Performance or readiness requirements. BIT provisions shall be incorporated in the controls to provide a GO/NO-GO indication of control readiness. These provisions may be in conjunction with, but not necessarily as part of, the fault isolation features. The operation of the performance or readiness test provisions shall be either a manually energized or continuous test mode of operation. Normal equipment performance may be interrupted during the manually initiated BIT sequence. Self-test of the BIT is not required.

3.4.2.3.2 Fault isolation. In conjunction with, but not necessarily as part of, the performance or readiness test features, there shall be features incorporated which provide fault isolation to the WRA.

3.4.2.3.3 MTTR (mean time to repair). For organizational maintenance, the MTTR shall be not greater than 15 minutes to remove and replace and verify operation of the replaced equipment. The control design shall be such that no ancillary test equipment is required for organizational level maintenance.

3.4.2.4 Intermediate level maintainability. Intermediate level maintenance may be performed with the assistance of separate test equipment. The time to verify performance at the intermediate level maintenance shall be not greater than 15 minutes.

3.4.3 Transportability. The controls shall be transportable by air, rail, truck, and ship when packaged in accordance with the requirements of Section 5.

3.4.4 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests, except that required field tests may be conducted under outside ambient conditions:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

a.	Temperature	Room ambient (25 ± 10°C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input power voltage	28 ± 0.5 vdc

3.4.4.1 Service conditions. The controls shall operate as required under any of the environmental service conditions or combination of these conditions as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

3.4.4.1.1 Sinusoidal vibration. The controls shall not require isolators or rear support and shall operate as required when subjected to sinusoidal vibration in accordance with test procedures of MIL-T-5422 to the requirements of Curve IIIB of MIL-E-5400.

3.4.4.1.2 Random vibration. The controls shall meet the random vibration requirements specified in 4.4.4.

3.4.4.1.3 Rain protection of console mounted equipment. The portion of the console mounted controls which are normally exposed when the equipment is mounted in a console shall meet the rain requirements of MIL-STD-810, Method 506.2, Procedure II.

3.4.4.1.4 Thermal conditions. The controls shall operate as required under the thermal conditions specified in MIL-E-5400, Class 2, without the requirement for auxiliary cooling.

3.4.4.1.5 Temperature shock. The control shall meet the temperature shock requirements of MIL-STD-810, Method 503.2.

3.4.4.2 Warm-up time. The time required for the controls to warm up prior to operation shall be kept to a minimum and shall be not greater than 10 seconds at any operating environment.

3.4.4.3 Electromagnetic interference (EMI). The control shall comply with MIL-STD-461 with Notice 3 for Class 1/A equipment except as modified in Tables III, IV, and herein. These requirements shall be met under all combinations of operating modes and frequencies of the control.

3.4.4.3.1 Conducted and radiated emissions limits. The conducted and radiated emissions of the control shall comply with the limits of the following requirements of MIL-STD-461:

a. CE01

b. CE02

c. CE03

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

d. CE04

e. RE01

f. RE02

3.4.4.3.2 Conducted and radiated susceptibility limits. The control shall comply with conducted and radiated susceptibility limits of the following requirements of MIL-STD-461:

a. CS01

b. CS02

c. CS06

d. RS01

e. RS02

f. RS03

The control shall exhibit no susceptibility as defined in 3.4.4.3.3 when subjected to an RS03 test to the limits of Table III. In addition, the control shall exhibit no susceptibility in accordance with the criteria of 3.4.4.3.3.b when subjected to signals at the levels of Table IV.

3.4.4.3.3 Susceptibility. The following criteria shall be applied to determine whether the control is susceptible:

a. Any failure of the control to comply with the performance requirements of 3.6 and 3.7 shall be considered an indication of susceptibility.

b. Any repeatable non-command change in the modes of operation or present frequencies of the control shall be considered an indication of susceptibility. This shall apply whether the failure is transient (momentary) or non-transient (permanent) in nature. In addition the inability to change radio frequency or mode of operation while exposed to the susceptibility signal be considered indication of susceptibility.

3.4.4.3.4 Ground plane interference. All control interface circuits shall be capable of specified performance when subjected to the following aircraft chassis noise between the WRA's.

a. Three volts rms from 320 Hz to 500 Hz. (Not to be greater than 150ma rms applied current.)

b. One volt rms from 500 Hz to 20 MHz. (Not greater than 150ma rms applied current for frequencies between 500 Hz and 50 kHz. Not to be greater than 1 watt (CW) from a 50-ohm source for frequencies above 50 kHz.)

c. ± eight volt pulses, 100 sec wide at 100 pps. (Not to be greater than 15a peak current spikes.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE III. RSO3 signals

Frequency (MHz)	Modulation				Power Density (mW/cm^2) Peak 5/	Power Density (mW/cm^2) Avg	v/m (ref) 6/
0.014-850 7/	AM 90% 1 kHz Sine Wave				-	.106	20
215-225	PAM 1/	20 usec 2/	PW 3/	200 pps 4/	.317	-	35
215-225	PAM	200 usec	PW	200 pps	.317	-	35
400-450	PAM	15 usec	PW	300 pps	.106	-	20
850-940	PAM	10 usec	PW	1000 pps	.106	-	20
850-940	PAM	125 usec	PW	300 pps	.106	-	20
1215-1365	PAM	10 usec	PW	3000 pps	.106	-	20
1365-2900	PAM	1 usec	PW	1000 pps	.106	-	20
2900-3100	PAM	3 usec	PW	1000 pps	1.6	-	78
2900-3100	PAM	70 usec	PW	200 pps	1.6	-	78
3100-9000	PAM	1 usec	PW	1000 pps	.106	-	20
9000-14000	PAM	0.2 usec	PW	1000 pps	.106	-	20
14000-18000	PAM	0.2 usec	PW	50000 pps	.106	-	20

1/ Pulse Amplitude Modulation.
2/ Microseconds.
3/ Pulse Width.
4/ Pulses Per Second.
5/ Peak Power Density for PAM signals is specified as the power density measured by an average power measuring device if the generator is run CW; average power density within the pulse envelope.
6/ Volts Per Meter equivalent shown for reference only.
7/ Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations and field strength shown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE IV. High level RS03 signals

Frequency (MHz)	Modulation				Power Density (mw/cm^2) 5/	v/m (ref) 6/	Scan Rate (sec) 7/	Dwell Time (msec) 7/
220 8/	PAM1/	10 usec2/	PW3/	200 pps4/	10.6	200	4	500
220	PAM	200 usec	PW	200 pps	10.6	200	4	500
425	PAM	15 usec	PW	300 pps	10.6	200	10	200
900	PAM	2 usec	PW	1000 pps	10.6	200	5	50
900	PAM	125 usec	PW	300 pps	10.6	200	5	50
1250	PAM	10 usec	PW	3000 pps	10.6	200	5	50
1350	PAM	5 usec	PW	3000 pps	10.6	200	5	50
3000	PAM	3 usec	PW	1000 pps	10.6	200	5	50
3000	PAM	70 usec	PW	200 pps	10.6	200	5	50
3600	PAM	1 usec	PW	1000 pps	10.6	200	5	50
5600	PAM	1 usec	PW	600 pps	10.6	200	5	50
9100	PAM	0.5 usec	PW	1000 pps	10.6	200	5	50
9375	PAM	0.2 usec	PW	1000 pps	10.6	200	5	50
13900	PAM	0.2 usec	PW	50000 pps	10.6	200	0	CONT
15500	PAM	0.2 usec	PW	50000 pps	10.6	200	0	CONT
33200	PAM	0.2 usec	PW	2000 pps	10.6	200	0	CONT

1/ Pulse Amplitude Modulation.
2/ Microseconds.
3/ Pulse Width.
4/ Pulses Per Second.
5/ Peak Power Density for PAM signals is defined as the power density measured by an average power measuring device if the generator is run CW; average power density within the pulse envelope.
6/ Volts Per Meter equivalent shown for reference only.
7/ Scan and Dwell refer to simulated antenna rotation. The signal shall be disabled except for the specified dwell. The signal shall be periodically enabled for the specified dwell time with the period equal to the specified scan time.
8/ Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations shown.

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

In addition to the susceptibility of 3.4.4.3.3, the radio shall maintain intelligible communications during a "talk thru" test with another radio set using a low level signal (not to be greater than 20" volts input to the receiver). The control shall be tested with a receiver-transmitter.

3.4.4.3.5 Conducted susceptibility test, interconnecting and signal leads. The unit shall exhibit no susceptibility, as defined by paragraph 3.4.4.3.3 when subjected to 100 mv rms 2 to 30 MHz coupled into its interconnecting and signal leads.

3.4.4.3.6 Grounding requirements. The grounding system between WRA's of the equipment shall be compatible and consistent with the aircraft grounding scheme specified in 3.4.4.3.6.1 through 3.4.4.3.6.3. All WRA's shall be grounded in such a manner as to prevent ground loops and common ground returns for signal and power circuits, provide effective shielding for signal circuits, minimize EMI, and to protect personnel from electrical hazards.

3.4.4.3.6.1 Chassis grounds. A ground wire of minimum length connected internally to the WRA chassis shall be provided at a pin on each primary power connector. No circuit shall be allowed to utilize this wire as its primary return.

3.4.4.3.6.2 Shielding grounding. A separate connector pin shall be provided for each aircraft wire shield unless suitable grounding is provided in external cable backshells. The connector shield pins shall be grounded to the equipment chassis inside the WRA adjacent to the connector mounting and by the shortest means practical.

3.4.4.3.6.3 Component grounds. All externally exposed metal parts, shields, control shafts, switch handles, connectors, and bushings, shall be grounded to the chassis.

3.4.4.3.7 Shielding gaskets. Shielding gaskets and similar devices for RF applications shall avoid the use of materials which represent severe dissimilar metal combinations with the interface.

3.4.5 Dimensions. The dimensions of the controls shall be as shown on Figure 1.

3.4.6 Weight. The weight of the control shall be not greater than 3.0 pounds.

3.4.7 Color. The front panel color of the control shall be lusterless black in accordance with FED-STD-595, Color No. 37038.

3.4.8 Nameplates and identification marking. Nameplate approval, and identification marking shall be in accordance with MIL-N-18307.

3.4.8.1 Decals, nameplates, and marking locations. All decals, nameplates, and marking shall be affixed to the controls as shown on Figure 1.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

3.4.8.2 Serial numbers. Serial numbers will be assigned by the procuring activity for the controls under contract or purchase order. The contractor shall serialize sequentially all controls which are delivered under contract

3.4.9 Input electrical power.

3.4.9.1 Operating power. The controls shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as specified in MIL-STD-704. The voltage supplied shall be 28 volts dc. The power required shall be not greater than 15 watts. The operating power requirements shall not include the power requirements of 3.4.11.2.

3.4.9.2 Lighting power. Input power for lighting of any lighted panel shall not require more than 5.0 watts at 5 or 28 volts (ac or dc).

3.4.9.3 Degraded performance. Degraded performance will be permitted for voltage transients with durations of 75 microseconds to 0.5 seconds during normal electrical system operation. Operation shall return to normal within five seconds with no resulting damage to the control.

3.4.9.4 Emergency voltage. During operation in the emergency steady state electric power condition specified in MIL-STD-704, the control shall remain safe and frequency readouts and control functions shall be maintained. The emergency power source will impose an interval of zero voltage at time of loss of primary power for a maximum period of 30 seconds, but shall return to performance as specified above after return to emergency power conditions in accordance with MIL-STD-704.

3.4.9.5 Undervoltage protection. The controls shall not be damaged by voltages below the minimums specified by MIL-STD-704 and shall resume normal operation when the voltage returns within normal limits.

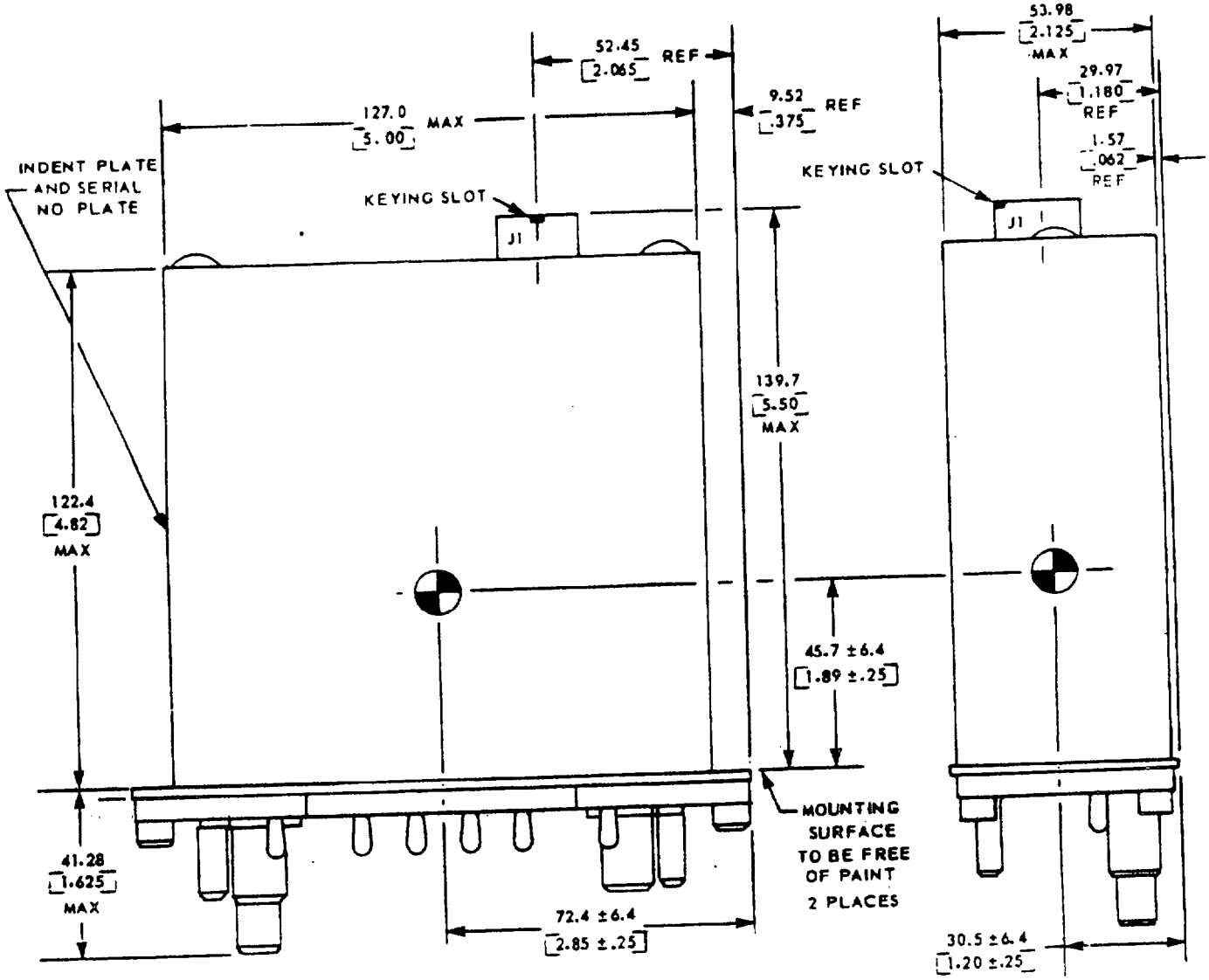
3.4.9.6 Reverse polarity protection. The control shall be protected such that it will not be damaged by application of reverse polarity voltages.

3.4.10 Manufacturing requirements. The control shall be manufactured in accordance with manufacturing processes, workmanship, and quality assurance provisions of MIL-STD-454, MIL-STD-275, MIL-P-55110, WS-6536, NAVMAT P-4855.1, and NAVMAT P-9492.

3.5 Performance. Unless otherwise specified herein, values set forth to establish the requirements of specified performance apply to performance under both standard and service and input power conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.5.1 Operation. The control shall provide the functions and capabilities specified herein. Additionally, the control shall be compatible with associated equipment listed in 6.5 to provide operations specified in 3.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.



DIMENSIONS ARE IN MILLIMETERS [INCHES]

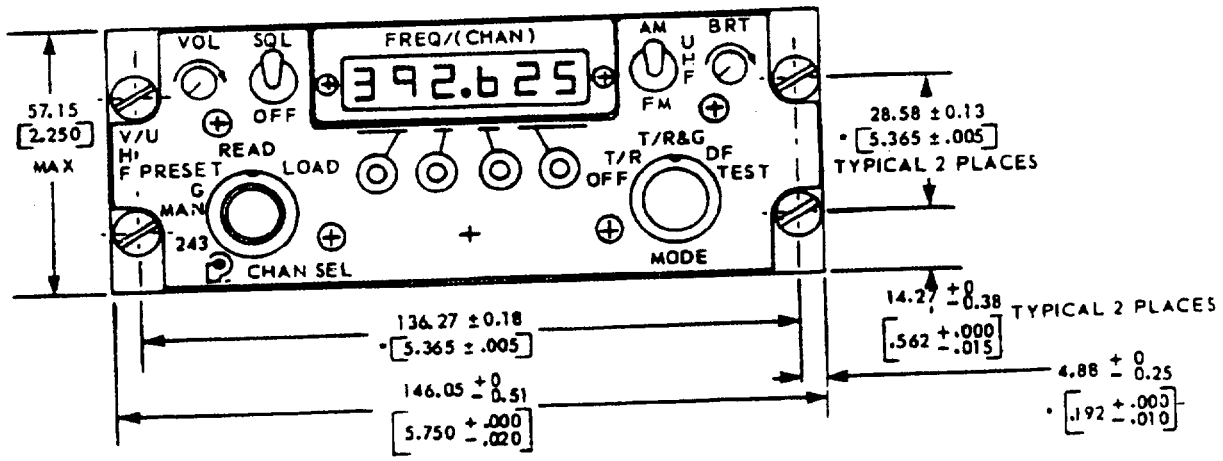


FIGURE 1. Radio Set Control C-10319A/ARC, C-10776A/ARC and C-11131/ARC

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

3.6 Detail requirements.

3.6.1 Function. The control shall provide remote control of the host receiver-transmitters in all modes specified.

3.6.2 Controls. Operating controls required to operate the host receiver-transmitter shall be located on the control/indicator assembly at the front of the control unit. The input and output frequency and channel information shall be in a serial digital format such that information transfer to and from the associated receiver-transmitter and indicators can be accommodated by two pairs of wires. Other control functions shall be accommodated by conventional wiring for the respective functions.

3.6.2.1 Operation modes.

a. OFF. In this mode, the control by discrete interface shall cause the internal dc power to be electronically switched/removed from all but the on/off circuitry of this control unit and the associated receiver-transmitter and remote indicator when used.

b. Main receiver-transmitter mode (T/R). The control shall cause the main receiver to be turned on and the guard receivers shall be in the off condition. The transmitter shall be in the ready (key up) condition.

c. Main receiver-transmitter guard mode (T/R+G). In this mode, the control shall cause both main and guard receivers to be turned on and the operator shall be able to hear the audio output of both receivers simultaneously. The guard receiver selected shall be that one which is in the frequency band of the main receiver frequency selected. The radio set shall have the same transmit-receiver capabilities as specified in the main receiver-transmitter mode.

d. ADF mode. In this mode, the main receiver-transmitter and guard receiver shall be operating as specified in 3.6.2.1c above, plus the radio set shall be capable of operation with the associated automatic direction finder equipment.

e. Built-in-test (BIT). In this mode, a performance check shall be applied to the radio to verify radio set operation. The GO/NO-GO results of the BIT shall be displayed by an indication on the front panel of the control.

3.6.2.2 Frequency select modes. The frequency select modes shall allow the operator to select emergency, guard, manual, or preset modes of frequency selection. In addition, provisions shall be made for the preset loading and preset read functions.

a. Emergency mode (243). In this mode, the radio shall be turned on and tune the transmitter and the main receiver to the 243.000 MHz (AM) guard frequency. When operating in this mode, all other controls shall be inoperative. This mode shall have precedence over other modes.

b. Manual mode. This mode shall allow the operator to rapidly and easily select any one of the possible operating frequencies by use of controls specified in 3.6.2.7. Any operating channel frequency selected shall be shown on the front panel using the display specified in 3.6.3. When operating in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

the manual mode, it shall be impossible to change or affect the operating frequency of the radio set by means of the preset frequency select control. The transmitter and receivers shall be disabled while changing frequencies.

c. Guard mode. In this mode, the radio shall automatically tune the main receiver and transmitter to the guard channel of the selected frequency band and shall readout the appropriate frequency on the display as specified in 3.6.3. The selected frequency band shall be considered to be the frequency band that incorporates the frequency or channel last selected on either the preset (if the radio was last used in the preset mode) or the manual frequency controls (if the receiver-transmitter was last used in the manual mode). When operating in this mode, the preset frequency selector control as well as the manual frequency selector controls shall be inoperative.

d. Preset mode. This mode shall allow the operator to rapidly and easily select, by only one control, 30 preset operating channels by use of the control specified in 3.6.2.8. The channel number (1 through 30) and the frequency of any operating channel selected shall be shown on the display as specified in 3.6.3. When operating in this mode, it shall be impossible to change the operating frequency of the radio set by means of the manual frequency selector controls. The transmitter and receivers shall be disabled while changing frequencies.

e. Preset read mode. This mode shall display the preset channel frequency to which the receiver-transmitter is tuned in lieu of the preset channel number. The frequency will be shown on the display as specified in 3.6.3. It shall be possible to select and display (but not load or change) any proper frequency in this mode. Changing and loading of the preset frequency shall be accomplished by using the preset load mode only.

f. Preset load. This mode shall be provided to transfer the selected frequency to the desired channel in the preset memory. The selected frequency for loading may be either the frequency last selected or either the manual or preset read mode. The position shall be designed in such a way as to prevent unintentional loading.

3.6.2.3 Improper frequencies. Under no conditions shall the radio tune to improper frequencies. These frequencies are:

a. Below 30.000 MHz

b. Between 88.000 through 107.975 MHz

c. Between 174.000 through 224.975 MHz

d. Above 399.975 MHz

In the frequency range from 108.000 MHz through 117.975 MHz where the receive mode is proper, the control shall not permit transmission.

3.6.2.4 Audio volume control. The audio volume control shall allow the operator to manually adjust the audio power output level. The control shall provide maximum output in the full clockwise position with attenuation increasing at least 48 dB when rotating in the counterclockwise direction to

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

approximately 95 percent of the rotation. In the full counter-clockwise position (100 percent), no audio shall be present. The attenuation change, in dB, shall be essentially linear with degree of rotation. No audio amplification shall be provided by the control unit.

3.6.2.5 Squelch control. The squelch control shall allow the operator to disable the squelch circuit on the main receiver only. In the off condition, the squelch circuit shall be disabled to allow receiving of main receiver signals regardless of carrier to noise characteristics.

3.6.2.6 Light dimmer control. A light dimmer control shall be provided on the front panel to adjust the illumination intensity of the display as specified in 3.6.3.

3.6.2.7 Manual frequency select controls. Control shall be provided on the front panel to manually select any of the radio frequencies.

3.6.2.8 Preset frequency select control. One selector control shall be provided on the front panel for selection of any one of the 30 available preset channels.

3.6.2.9 AM/FM select. In all bands except the VHF band the AM/FM modes shall be automatic to provide:

a. FM in the 30 to 88 MHz band

b. AM in the 108 to 156 MHz band

c. FM in the 156 to 174 MHz band

For the UHF band an AM/FM select switch shall be easily accessible to choose the AM or FM mode. This switch shall be configured such that unintentional switching is precluded.

3.6.2.10 Take control function. A pin shall be provided on each control which shall, when grounded, disable that control and allow another control to operate the radio unit (see EI-699, Drawing 18, Note 6). The transmitter-receivers shall be disabled while changing frequencies when this function is selected.

3.6.3 Frequency/channel display. A six digit frequency display shall be provided on the front of control to display the operating radio frequency. A decimal shall also be lighted to identify the MHz position of three significant digits (i.e., 225.000). For preset channel number display, only one or two digits (for preset channel number 1 through 30) shall be lighted with the remaining digits unlighted. The display shall be arranged horizontally in a centrally located position on the front panel of the control. The display shall be such that the digits can be easily read from a distance of 2.0 feet at any angle up to at least 30 degrees from the display centerline perpendicular to the surface of the front panel. Provisions shall be made for easy removal and replacement of the display digits without loosening or removing the equipment front panel or control knobs. Additionally, the control shall provide serial data output to operate equivalent displays of a panel mount radio set remote indicators.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.6.4 Electrical connections. Connections to external circuits shall be provided on receptacles as specified in Table V.

TABLE V. Electrical receptacles and functions

Reference Designation	Receptacle Type	Function
J1	Bendix TVP02RW-15-35P (MIL-C-38999, Series III)	Power/Control

3.6.5 Receiver-transmitter interface. The control shall communicate with the host receiver-transmitter, one or more indicators, and secure voice equipments through a combination of parallel control lines and a serial data bus.

3.6.5.1 Parallel control lines. The control shall interface with parallel control lines via J1, as specified electrically and functionally in Tables VI and VII.

3.6.5.2 Serial data bus. The ARC-182 serial data bus shall be used to communicate between the control, the host receiver-transmitter and optional indicators. The serial bus consists of four lines - two each for data and clock. The electrical characteristics of these lines are shown in Table VI (pins J1-10, 18, 19, and 20). See 3.7 for further definition of the serial interface.

CAUTIONARY NOTE

ARC-182 SERIAL CONTROL INTERFACE SPECIFICATION

All devices connected to the serial data bus should read and decode all data information. Further, the device should decode command word information to compare data sent versus data responses, to insure that no inconsistencies have occurred, either as a result of false clock pulses, bit transitions or external interference.

3.6.5.3 Single control, two Radio RT-1250A/ARC Receiver-Transmitter system. A single control shall have capability for operating two Radio RT-1250A/ARC Receiver-Transmitters primarily via the serial data bus. Frequency change information shall be accepted by one receiver-transmitter at a time and as designated by a separate external switch and one shot pulse generator circuit. One receiver-transmitter shall accept the control generated input following a 30 millisecond pulse and the other shall accept an input generated after an 80 millisecond pulse. Detailed interface information for this system is covered in 3.7.

3.6.6 Preset channel memory device. The memory device shall be a nonvolatile type such that power off, power line interruptions or any combination of environmental conditions will not cause a loss or change of preset channel or preset.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.6.7 Audio interface. The control shall accept unbalanced, unattenuated audio from the host receiver-transmitter at input levels up to 1 watt rms. The input impedance shall be 100 ohms minimum when the output is loaded with the specified impedance. The control shall provide balanced audio output at either 600 ohms or 150 ohms, not both simultaneously. The audio output shall be modified by the position of the volume control specified in 3.6.2.4 with a maximum output no less than 3 dB below the audio input over the frequency range of 400 to 3,000 Hz.

TABLE VI. Remote control interface characteristics
J1 - power/control connector 1/

Pin No.	Pin Function	I/O	Characteristics
1	DC Power In	I	+28 volts dc power in accordance with MIL-STD-704, 15 watts maximum.
2	Panel Lamp Power (5V)	I	5 volts ac or dc, 5 watts maximum use either pin 2 or pin 14, not both.
3	DC Power Return	I	System ground.
4	Unattenuated Audio (Low)	I	Voice audio from receiver-transmitter. Low side may be grounded.
5	Unattenuated Audio (High)	I	6.5 ± 1.0 VRMS into 1000 ohm load.
6	Attenuated Audio (High)	O	150 ohm output, P_{out} (audio) ≥ P_{in} -3dB
7	Attenuated Audio (Low)	O	Audio common
8	Open (Reserved)		
9	Spare		
10	Data (HI)	I/O	"1" = 4.8 VDC ± 0.5 volt dc; "0" = 0.0 volt dc ± 0.5 volt dc.
11	Spare		
12	Open		Jumper to pin 34 for "have quick" (HQ) compatibility.
13	Signal Ground	O	Ground for shields

1/ This table indicates nominal characteristics. For specification limits, see the appropriate paragraph.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE VI. Remote control interface characteristics
J1 - power/control connector - Continued

Pin No.	Pin Function	I/O	Characteristics
14	Panel Lamp Power (28V)	I	28 volts ac or dc, 5 watts maximum, Use either pin 2 or pin 14, not both.
15	Open		Jumper to pin 28 for HQ compatibility.
16	Attenuated Audio Out (High)	O	600 ohm output P_{out} (Audio) $\geq P_{in}$ -3 dB
17	Take Control	I	Open - control in command of receiver-transmitter; Ground - other unit in command.
18	Data (LO)	I/O	See pin 10, line is at "1" when not active.
19	Clock (HI)	I/O	"1" = 4.8 VDC $\pm$ 0.5 VDC; "0" = 0.0 VDC $\pm$ 0.5 VDC.
20	Clock (LO)	I/O	See pin 10, line is at "1" when not active.
21	Panel Lamp Common	I	Power return for panel lamp power.
22	SMC 2	I	Ground inhibits handshake on bus; open enables response. Line shall be grounded along with pin 17, when two receiver-transmitters are connected to the same bus.
23	SMC 3	I	"Low" = 0V $\pm$ 0.5 VDC.
24	Open (Reserved)		
25	SMC 1	I	Spare system mode control line.
26	ON/OFF	I/O	0 to +7 volts dc turns system on, open circuit turns system off.
27	Open (Reserved)		
28	Open		Jumper to pin 15 for HQ compatibility.
29	Open (Reserved)		
30	Open (Reserved)		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE VI. Remote control interface characteristics
J1 - power/control connector - Continued

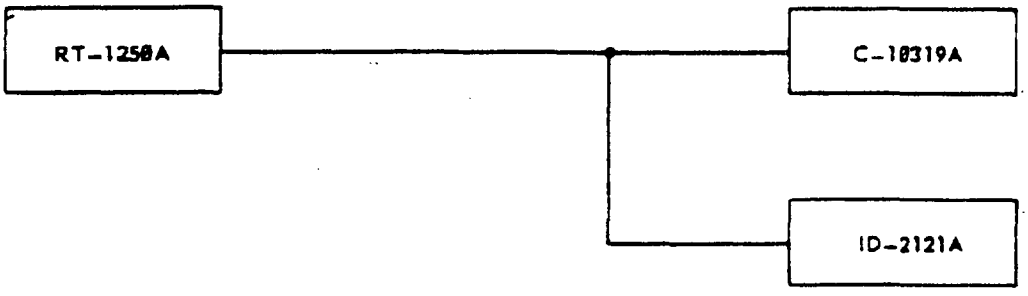
Pin No.	Pin Function	I/O	Characteristics
31	Open		Jumper to pin 32 for HQ compatibility.
32	Open		Jumper to pin 31 for HQ compatibility.
33	Open (Reserved)		
34	Open		Jumper to pin 12 for HQ compatibility.
35	Open (Reserved)		
36	Guard Precedence		Ground to select 243 MHz.
37	Open (Reserved)		

TABLE VII. System mode control (SMC)

System	SMC 1	SMC 2
One Radio RT-1250A/ARC Receiver-Transmitter and One Control	Open	Open
Two Radio RT-1250A/ARC Receiver-Transmitters and One Control	Open	Ground
Spare		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.7 ARC-182 serial control. Receiver-transmitters, remote controls, and remote indicators shall communicate by way of a serial data bus, the data shall be transmitted synchronously using 5 volt differential signals. Clock and data shall be carried on two twisted shielded pair cables. The radio sets and controls shall be capable of receiving and transmitting data on the serial data bus. The indicators shall be capable of receiving data only. Figure 2 illustrates a typical ARC-182 system.



Note: The C-10319A/ARC and ID-2121A/ARC are typical units; the C-10319A/ARC is listed in the diagram and text; this specification applies equally to the C-10776A/ARC and C-11131/ARC, for all characteristics specified herein.

FIGURE 2. Typical ARC-182 system configuration

3.7.1 Data signals. Data is transferred by DATA HI and DATA LO signals. DATA HI is the true state of the data being transferred, while DATA LO is the inverse. A differential synchronous clock shall be sent along with the data signals. CLK HI (clock high) shall be normally low with its true or clock state being high. CLK LO shall be normally high with its true or clock state being low. Timing diagrams are shown on Figure 3. Data shall be latched into the receiving unit on the negative going edge of CLK HI.

3.7.2 Signal characteristics. CLK HI-CLK LO and DATA HI-DATA LO shall be differential pairs. Each line shall have a signal to ground low level amplitude of 0.5 volt + 0.5 volt and a high level amplitude of 4.8 volts + 0.5 volt. These signals shall be measured as shown on Figure 4. All ARC-182 control units shall drive a maximum of 100 feet of twisted shielded pair cable (maximum of 30 PF/FT line to ground). Clock and data signals shall have a maximum rise time (10 percent - 90 percent) of 40 microseconds.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

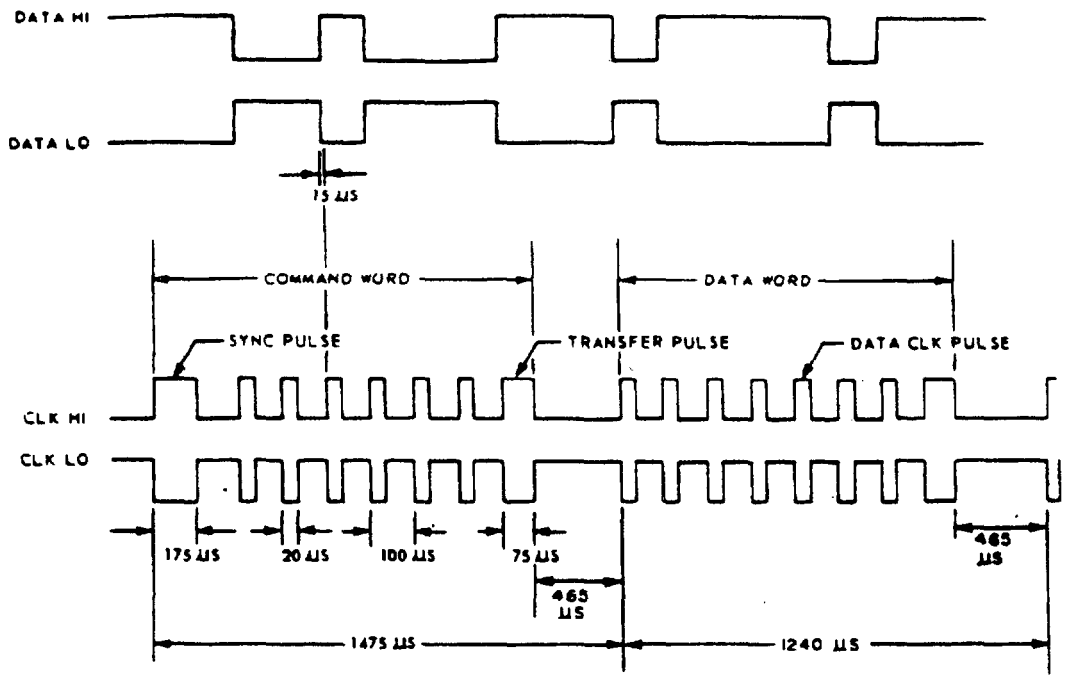


FIGURE 3. Serial data tuning

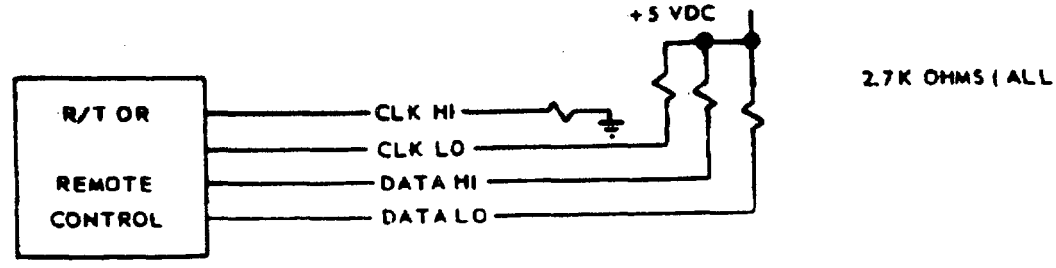


FIGURE 4. CLK HI, CLK LO, DATA HI and DATA LO signal level measurements

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

3.7.2.1 Clock types. Three types of clocking pulses shall be sent on the ARC-182 serial data bus. Each of these clock signals shall have a different pulse width (see Figure 3).

3.7.2.1.1 Synchronization pulse. The synchronization pulse shall be sent at the beginning of each message to signal the start of a data message. Data lines do not carry valid information during a synchronization pulse. The synchronization pulse shall have a pulse length of 175 microseconds ± 10 microseconds.

3.7.2.1.2 Data clock pulse. The data clock pulse shall be used to clock data into the receiving unit. The data lines shall carry valid information during the data clock pulse. The data clock pulse shall have a pulse length of 20 microseconds ± 1 microsecond.

3.7.2.1.3 Transfer pulse. Every command and data word shall end with a transfer pulse which signals the end of a word. The receiving unit can use the transfer pulse to latch data on the parallel outputs if a serial to parallel register is used. The data lines carry valid information during the transfer pulse. The transfer pulse shall have a pulse length of 75 microseconds ± 4 microseconds.

3.7.2.2 Data message. Each data message shall begin with a receive command word followed by data words (see Figure 5).

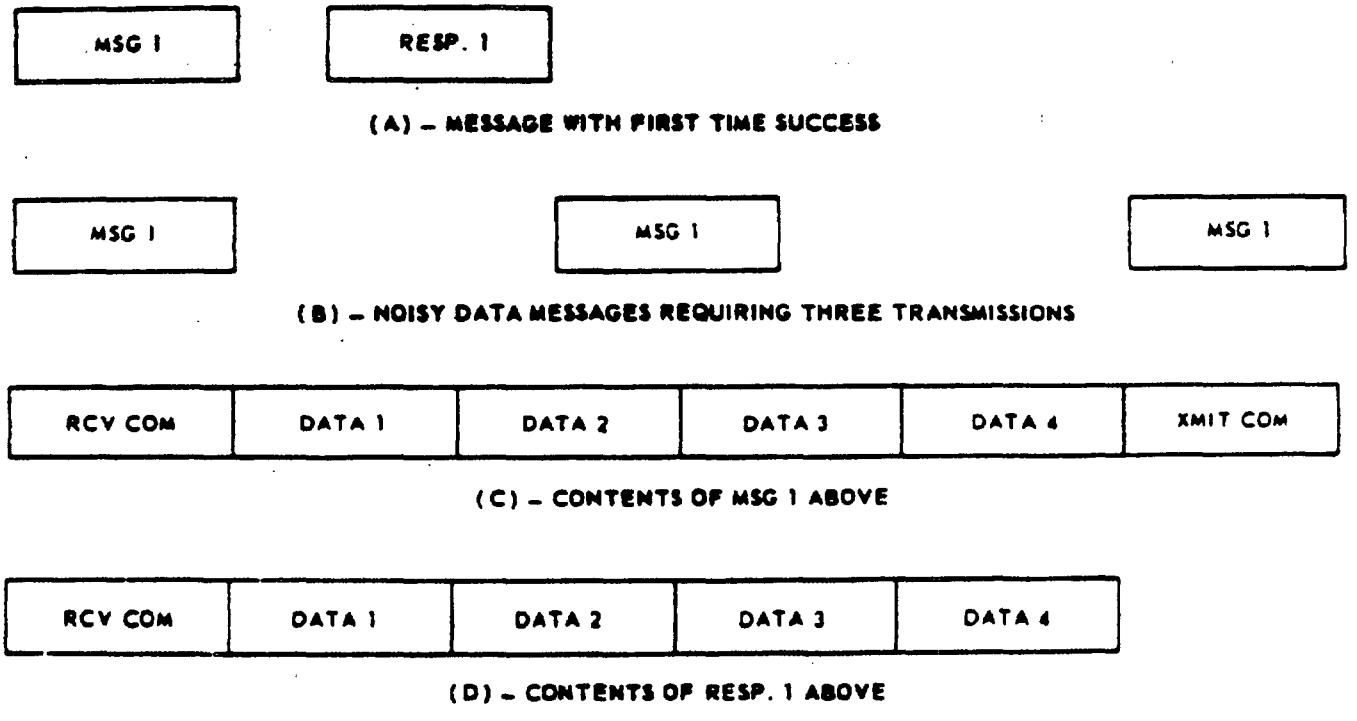


FIGURE 5. Data messages

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

3.7.2.2.1 Remote control in command. A transmit command word shall be sent after the last data word which commands the receiver-transmitter to echo back the data most recently received. The echoed data is compared against the data sent. If the echoed data agrees exactly with the data sent, no further data shall be sent. If the echoed data does not agree with the data sent, the remote control shall send the data again. The data shall be sent up to three times in attempting to achieve agreement. If the final message received by the receiver-transmitter contains valid mode and frequency information, the receiver-transmitter shall update. If invalid mode or frequency information is contained in the third attempt, the receiver-transmitter shall remain on the last valid data update. The remote control shall display the data sent.

3.7.2.2.2 Receiver-transmitter in local command. With the receiver-transmitter in local command, the data shall be sent three times by the receiver-transmitter. The remote controls and indicators do not have echo back capability and shall update to the last data message received.

3.7.3 Word formats. Two general types of words shall be sent on the ARC-182 data bus; command word containing message overhead information and a data word containing frequency, mode, or built-in-test information.

3.7.3.1 Command word. The command word format and bit definition shall be as shown on Figure 6. The command word shall be preceded by a synchronization pulse. All command words shall be seven data bits in length.

3.7.3.1.1 First data word address. The first bits sent (B_0 - B_2) shall contain the starting address where the first data word shall be stored. All data words shall be stored in a specific order in memory within each unit. Having knowledge of the starting address, the data word contents can be determined and store in the correct location. Data words shall always be sent from sequential locations.

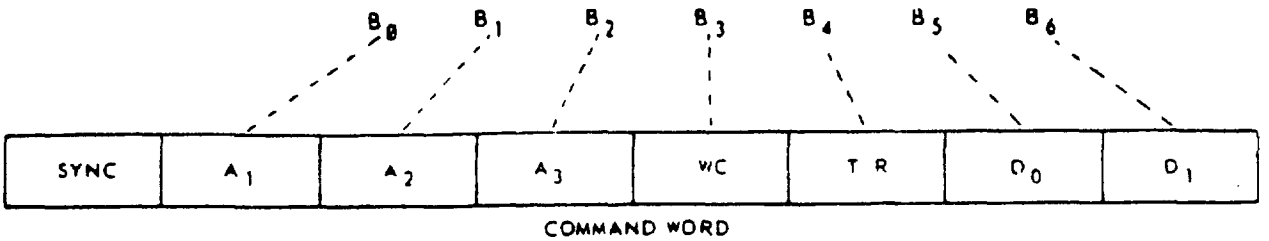
3.7.3.1.2 Word count bit. The word count bit (B_3) shall specify how many data words will follow a receive command word. This can be selected as two or four data words. A transmit command shall never have data words following it.

3.7.3.1.3 T/R bit. Bit time five (B_4) shall be the T/R bit. This bit shall be used to indicate to the receiving unit whether it should receive or transmit data.

3.7.3.1.4 Unit address. The unit address (B_5 and B_6) shall define which units should receive and update to data on the bus. Presently in the ARC-182 radio system, there are four unit addresses. The unit addresses used by each control unit, when they are in local and remote command, are listed in Tables VIII and IX.

3.7.3.2 Data words. Information regarding mode, frequency, and built-in-test results shall be contained in data words. These data words along with respective definitions are listed in Table X. All data words shall be eight bits in length. The data word starting address are shown in Table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.



COMMAND WORD

TRANSMIT MESSAGE CONSISTS OF ONE COMMAND WORD

RECEIVE MESSAGE CONSISTS OF ONE COMMAND WORD AND TWO OR FOUR DATA WORDS

COMMAND WORD

A_1, A_2, A_3 = FIRST DATA WORD ADDRESS (B_0, B_1, B_2)

WC = DATA WORD COUNT (B_3)

B_3	WC
0	2
1	4

B_4 T/R BIT 1 = TRANSMIT

D_0, D_1 = UNIT ADDRESS (B_5, B_6)

B_5	B_6	RECEIVING TERMINAL
0	0	ALL CONTROLS, RTS AND INDICATORS
0	1	ALL C-10319 CONTROLS
1	0	PANEL MOUNT RADIO OR RT1
1	1	RT_2

FIGURE 6. Command word format

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE VIII. Transmit command word - unit address

Local Command	
Receiver-Transmitter 1	10
Receiver-Transmitter	2-11
Remote Control Unit	01

TABLE IX. Receive command word unit address

Units Which Respond to Unit Address	Unit Address B_5 B_6
All units	00
Remote Control Unit	01
RT_1 (Panel Mount)	10
RT_2	11

TABLE X. Data word format

Mode Word		
B_0 = 243 B_1 0 = Load B_2 0 = Sq. Disable B_3 1 = VHF FM		Note: B_0 = First bit sent. B_7 = Last bit sent.
B_4	B_5	Mode
0	0	T/R
1	0	T/R+G
0	1	Test
1	1	DF

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE X. Data word formats - Continued

B_6	B_7	Freq. Mode
0	0	Guard
1	0	Manual
0	1	Preset
1	1	Preset Read

Channel Word		
B_0	LSB	Binary Channel Number
B_1		
B_2		
B_3		
B_4	MSB	
B_5	Spare	
B_6	AV-8B	1=AV-8B Mode
B_7	T/R	1=XMIT

Frequency Words		
	Freq. 1	Freq. 2
B_0	1 MHz	100 KHz
B_1	2 MHz	200 KHz
B_2	4 MHz	400 KHz
B_3	8 MHz	800 KHz
B_4	10 MHz	25 KHz
B_5	20 MHz	50 KHz
B_6	40 MHz	100 MHz
B_7	80 MHz	200 MHz

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE XI. Starting address

Starting Address B_2 B_1 B_0	Data Word
$111A_0$	Mode Channel
$110A_0$	Frequency 1 Frequency 2
$101A_0$	Bit Results II Bit Results I
$100A_0$	Bit Results II Bit Results I

3.7.4 Bus activity. Normally, the ARC-182 data bus is quiet, with the receiver-transmitter in local command, frequency and mode information shall be sent only when there is a change in operational frequency or mode by manipulation of front panel controls (or MIL-STD-1553 mux bus on receiver-transmitters so equipped). Only the unit in command shall be capable of initiating bus traffic. When a control unit is not in command, all front panel switches shall be disabled and shall not effect operation of the receiver-transmitter. All data messages initiated by ARC-182 units shall be of the same format (see Figure 6 and Tables VIII, IX and X). Each message shall begin with a receive command word followed by data words. Messages shall always end with a transmit command word. This transmit command tells the unit to echo back the data it just received. When the receiver-transmitter is in remote command, it shall wait 18 milliseconds for a response. If no response is received before the timeout, the receiver-transmitter shall assume the echo compared and update to the new frequency.

3.7.5 Built-in-test (BIT).

3.7.5.1 Constant monitoring built-in-test. A constant monitoring built-in-test message shall be sent by the remote control when it is in command. This message is intended to periodically (0.5 second period) check the receiver-transmitter and verify that there are no constant monitoring faults. The constant monitoring message shall start with a transmit command from the remote control. The starting address of the command word shall point at the built-in-test results. The receiver-transmitter, in response to this command shall transmit a receive command followed by built-in-test results.

a. Control faults. Faults in the C-10319A/ARC Control that are detected by continuous/on-line built-in-test (BIT) shall be indicated by blanking the control display and lighting the decimal point.

b. Receiver-transmitter faults. Display of performance/readiness condition of the receiver-transmitter operating in normal mode is dependent on which unit is in command. If the control is in command, it requests constant monitoring BIT results from the receiver-transmitter every 0.8 seconds (0.5 seconds, nominal). This is done with a unique serial message consisting of only a transmit command (no data words) with bits B2, B1, and B0 = or 000

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

(i.e., pointing to BIT results address). The receiver-transmitter shall respond with receive message bit data words. The resulting display action in the control shall be as specified in Table XIII. Display action when the receiver-transmitter is in command is also shown in Table XIII. Note that the control monitors the serial bus clock line at all time for display of fault conditions which render the data lines inoperative.

3.7.5.2 Initiated (off line) built-in-test.

a. Control faults. Detected faults in the C-10319A/ARC control are indicated by the lighted decimal point, with other characters blanked.

b. Receiver-transmitter faults. Off-line BIT of the receiver-transmitter is indicated from the control by selecting the test mode and sending the proper mode message to receiver-transmitter. Data messages received from the receiver-transmitter are displayed as specified in Table XII.

TABLE XII. Display of receiver-transmitter performance/readiness

Serial Fault Messages			Unit in Command	
BIT Results (Hex)		Clock Lines	C-10319( ) Control 5/	Receiver-Transmitter
Word 1	Word 2			
FF	XX 2/	Normal	----.----1/	----.----
88	FF	Normal	No Change	No Change
No Response		Normal	----.----	N/A
----	----	Inverted, Static 3/	----.----	----.----
----	----	Toggling 4/	----.----	----.----

1/ A dash ("-") in any display format indicates an unlighted character.

2/ Symbols A, B, and C represent any number from 0 to 9. An X symbol means "don't care".

3/ When the serial clock lines invert and do not change further, the control shall display the fault as shown. (Normally, clock HI is low and clock LO is high, when not active).

4/ When the serial clock lines toggle at a slow rate (clock HI is low 50 to 225 millisecond, then high for 70 to 310 millisecond), the control shall display faults as shown.

5/ When control is in command, it shall request BIT results from receiver-transmitter every 0.8 seconds.

Source: https://assist.dla.mil -- Downloaded: 2022-2640-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE XIII. Display of receiver-transmitter off-line rest results

Serial Fault Messages			Unit in Command	
BIT Results (Hex)		Clock Lines	C-10319( ) Control 5/	Receiver-Transmitter
Word 1	Word 2			
FF	XX 2/	Normal	----.----1/	----.----
88	FF	Normal	No Change	No Change
88	88	Normal	888.888	888.888
FA	BC 2/	Normal	-A- B-C	-A- B-C
(Example) F1	32	Normal	-1- 3-2	-1- 3-2
No Response		Normal	-1- 5-7	N/A
----	----	Inverted, Static 3/	-5- 5-5	----.----
----	----	Toggling 4/	-1- 1-2	----.----

1/ A dash ("-") in any display format indicates an unlighted character.

2/ Symbols A, B, and C represent any number from 0 to 9. An X symbol means "don't care".

3/ When the serial clock lines invert and do not change further, the control shall display the fault as shown. (Normally, clock HI is low and clock LO is high, when not active).

4/ When the serial clock lines toggle at a slow rate (clock HI is low 50 to 225 millisecond, then high for 70 to 310 millisecond), the control shall display faults as shown.

5/ All off-line tests initiated from the control shall begin with display set ----.---- until new BIT results are received.

3.7.6 Single control unit, two RT-1250A/ARC system. In the AV-8B and possibly other aircraft systems it is necessary that one C-10319A/ARC or equivalent, control operate two Radio RT-1250A/ARC Receiver-Transmitter. Normally the receiver-transmitter (RT-1250A/ARC) is operated either from the MIL-STD-1553 Multiplex Bus or from a remote control without the need to share the control with similar Radio RT-1250A/ARC Receiver-Transmitter. In the AV8B the two receiver-transmitters are normally operated from the MIL-STD-1553 System with an alternate need for operating the two radios from one control. The two radios must be operated simultaneously with change commands directed to only one radio at a time.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

3.7.6.1 Effectivity designation. For switching to C-10319A/ARC or equivalent control, the receiver-transmitter pin 25 is grounded and the control pin 17 is open circuited by a separate aircraft system switch. The capability for directing commands to the number 1 or number 2 receiver-transmitter requires an external one shot pulse generator. By the acceptance coding connections to the receiver-transmitter J2 connectors, one radio will accept commands following a 30 millisecond pulse and the other radio will accept commands following an 80 millisecond pulse. The connections and functions associated with this capability are illustrated on Figure 7.

3.7.6.1.1 Operational description. After the system has been under the MIL-STD-1533B control and transferred to C-10319A/ARC, or equivalent control, the radios will remain at the previously assigned frequency and operating condition until changed by the C-10319A/ARC. When the radios are already operating, the C-10319A/ARC may be powered down to conserve power. The Radio RT-1250A/ARC Receiver-Transmitter is setup from the C-10319A/ARC Control as follows:

a. The control power shall be turned on.

b. Desired mode and frequency shall be selected via the front panel knobs and switches.

c. The external "send data" switch shall then be toggled.

As indication on Figure 7, toggling the number 1 radio position will cause a 30 millisecond pulse to be generated, whereas toggling the number 2 radio position will cause an 80 millisecond pulse to be generated. The receiver-transmitter message acceptance is determined by the wiring to J2 pin 7. The timing and wave shape forms required for the "send data" pulses are as shown on Figure 8. After the change data has been sent by the control and a valid response sent by the selected receiver-transmitter, the control shall display the receiver-transmitter's designated number for approximately 1.5 seconds. After this period the display of the control shall return to normal. If the control does not receive a valid response, its display shall not change.

3.7.6.2 External systems. Operation of two RT-1250A/ARC Receiver-Transmitters from one control shall require an external pulse generating circuit. A switch shall also be necessary to generate the proper pulse and initiate frequency/mode update of receiver-transmitter number 1 or receiver-transmitter number 2. The C-10319A/ARC monitors the SMC-3 line for pulse input. The pulse duration shall determine which receiver-transmitter is to be updated. A 30 millisecond pulse updates receiver-transmitter number 1 and 80 millisecond pulse updates receiver-transmitter number 2. The pulse generator characteristics required are shown on Figure 8. The system wiring for the pulse generator and association connections is illustrated on Figure 7. Grounding of control connector pin 22, SMC-2, inhibits updating except for command via pin 23, SMC-3. The RT-1250A/ARC receivers shall respond to 30 millisecond and 80 millisecond commands as controlled by the respective address wires. Grounding of J2 pin 1, Address A0, of receiver-transmitter number 1 and open circuit of J2, pin 1, address A0 of receiver-transmitter number 2 provides this differentiation. J2 pin 7, Address A1, of both RT-1250A/ARC radios and pin 22, SMC-2 of the control, are grounded to complete the coding for this mode of operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

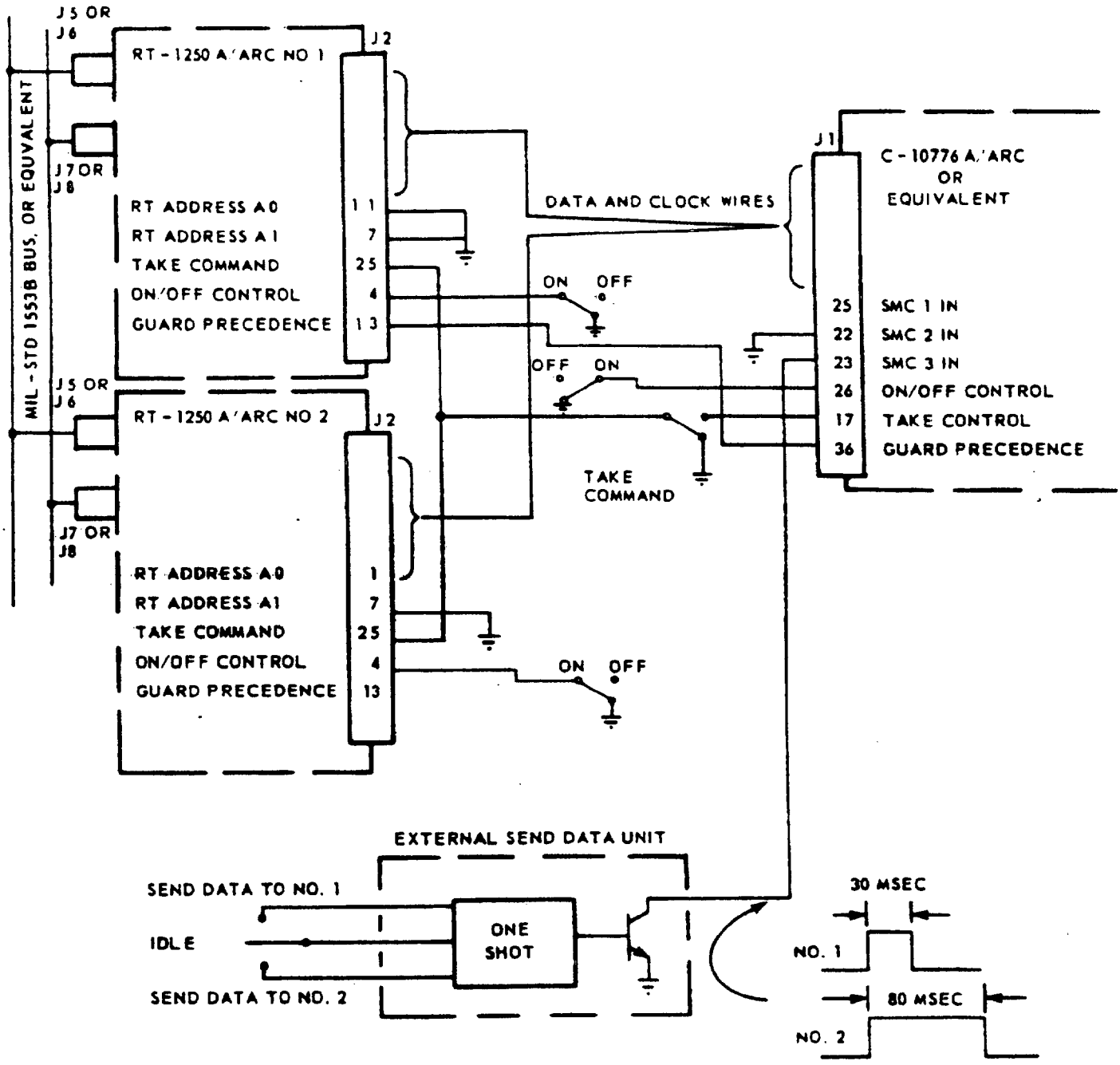


FIGURE 7. Dual RT-1250A/ARC, single C-10776A/ARC, or equivalent, unique interconnect characteristics

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

PULSE WAVESHAPE REQUIREMENTS:

RISE TIME (t_r)	100 MICROSECONDS
FALL TIME (t_f)	100 MICROSECONDS
SIGNAL LEVEL	"HIGH" OPEN (REF. TO +12 VDC IN RT-1250/ARC)
	"LO" 0 ± 5.0 VOLTS
PULSE WIDTH:	A: FOR RT-1250/ARC NO. 1 30 MS ± 3 MS
	B: FOR RT-1250/ARC NO. 2 80 MS ± 3 MS

GENERATOR OUTPUT IMPEDANCE ROUT (MAX) = 1 K OHMS AT 0.5 V

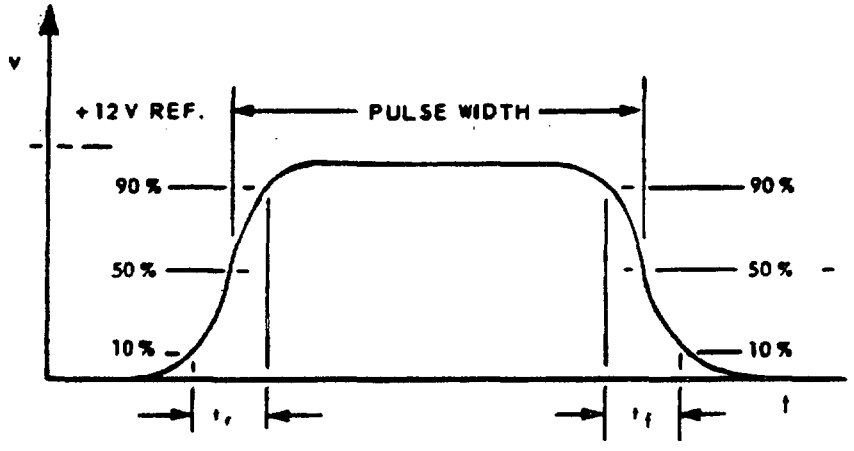


FIGURE 8. External pulse characteristic requirements

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

3.7.6.2.1 Alert mode. Separate power switches shall be provided for the RT-1250A/ARC radios and control for proper operation in the "Alert mode". This allows the control to be powered up independent of the radio. The control can also be powered down after use to reduce power drain. The RT-1250A/ARC radios will remain fully operational on the last frequency and mode commands after power is removed from the control.

3.7.6.2.2 Take command switch. Both the MIL-STD-1553 input and control input may be connected to the RT-1250A/ARC radio and be active simultaneously, however, effectivity shall be determined by the "take command" switch. No irregularities shall occur or change of operation shall result from the source not in command. Upon change of command from the control to MIL-STD-1553 or vice versa, the last operational mode and frequency shall not change until an update is issued by the unit in command.

3.7.6.3 Built-in-test (BIT). BIT can be conducted from the control on one receiver-transmitter at a time. BIT results shall be displayed on the control when it is in control. Initiation of BIT on receiver-transmitter number 1 shall result in the receiver-transmitter designated number being displayed, then followed by a GO/NO-GO display of the results for receiver-transmitter number 1 only. The same shall apply for receiver-transmitter number 2.

3.7.6.4 243 mode. The emergency mode, labeled 243 on the control designates receiver-transmitter operation on the guard frequency. When this mode is selected the receiver-transmitter number 1 shall be tuned to the 243.000 MHz guard frequency. If a mode change is made at the control, receiver-transmitter number 1 shall remain at 243.000 MHz until a send data pulse is initiated.

3.7.6.5 Guard precedence. Connection of the discrete guard precedence input of the control is optional in the AV-8B system. The guard precedence line can be conducted to the RT-1250A/ARC radios; however, it is recommended that only one radio be connected for guard precedence at any time.

3.8 Safety. Safety requirements shall be in accordance with MIL-E-5400 except resistance from ground to chassis shall be 20 milliohms maximum.

3.8.1 Human performance/human engineering. The requirements of MIL-STD-1472 and MIL-H-46855 shall be used as design guides for the controls and all components therein.

3.9 Bonding. Bonding shall be in accordance with MIL-B-5087.

3.10 Workmanship. Workmanship for the controls shall conform with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Standard conditions. Unless otherwise specified herein, all inspections shall be performed with conditions specified as follows:

a.	Temperature	Room ambient (25 $\pm$ 10°C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input Power	28 $\pm$ 0.5 volt dc

4.4 First article inspection. First article inspection shall be conducted by the contractor on controls representative of the production control to be supplied under the contract. First article tests shall be accomplished under the authorized test procedures (see 4.6). No first article inspections shall be conducted prior to acceptance of the first article test procedure by the procuring activity. First article inspection shall consist of all tests described in 4.4.1.

4.4.1 Scope of first article tests. First article tests shall include all tests in the approved test procedure to determine that the control meets all the requirements of this specification and the contract. First article tests shall include environmental tests in accordance with the procedures of MIL-T-5422 including temperature shock, a random vibration test as specified in 4.4.4, and an electromagnetic compatibility test in accordance with MIL-STD-461 and MIL-STD-462, as modified in 4.4.3. The first article test may also include a reliability qualification test.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:51Z
Check the source to verify that this is the current version before use.

4.4.2 First article approval. Approval of the first article sample will be by the procuring activity upon completion of all tests (see 6.2.2).

4.4.3 Electromagnetic interference tests. Compliance with the requirements of 3.4.4.3 shall be demonstrated by tests. These tests shall be performed in accordance with MIL-STD-462 as modified herein.

4.4.3.1 EMI Test Plan. The contractor shall perform EMI tests in accordance with a contractor prepared EMI Test Plan. The Test Plan shall describe all EMI tests required to demonstrate compliance with the requirements of 3.4.4.3, and shall address all anomalies, inconsistencies, and omissions (if any). The Test Plan shall detail a combination of operating modes and frequencies to be tested. The combination shall represent all operating modes of the control and at least four frequencies. The Test Plan shall be acceptable to the procuring activity prior to the commencement of EMI testing (see 6.2.2).

4.4.3.2 Ground plane interference. The contractor shall prepare a procedure to demonstrate that the control complies with the requirements of 3.4.4.3.4. The procedure shall be included in the EMI Test Plan (see 6.2.2).

4.4.3.3 Conducted susceptibility test, interconnecting and signal leads. In addition to the requirements of MIL-STD-461 the following RF susceptibility test shall be conducted. No malfunction, undesirable response, or change in indication beyond the tolerance given in the equipment specification or approval test plan shall be produced in any equipment when the RF signal levels listed in Table XIV are coupled into the interconnecting wiring.

TABLE XIV. Susceptibility test signals

Frequency	Required Signal Levels In One Turn Loop
2 to 30 MHz	100 MV RMS

The RF signal shall be modulated 80 percent with 400 Hz and with that frequency and the type of modulation to which the test sample is most susceptible. This requirement applies to all interconnecting wires or cable between component parts of subsystems, and to wires and cables connected to other equipment. The RF signal shall be measured in the one-turn loop with an RF voltmeter as shown on Figure 9. The signal generator shall have an output impedance of 50 ohms or less and a maximum open-circuit voltage of not less than 3 volts rms. The test procedure shall be as follows:

a. With the current probe clamped around the wire bundles under test, set the signal generator for maximum output and slowly scan from 2 MHz to 30 MHz.

b. At those frequencies where a response is noted, move the current probe along the wire for maximum indication. Reduce the signal generator output and determine the threshold of susceptibility of the wire bundle. When a response is found on an interconnecting wire bundle, probe each wire in the bundle separately to determine the susceptible wire(s).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

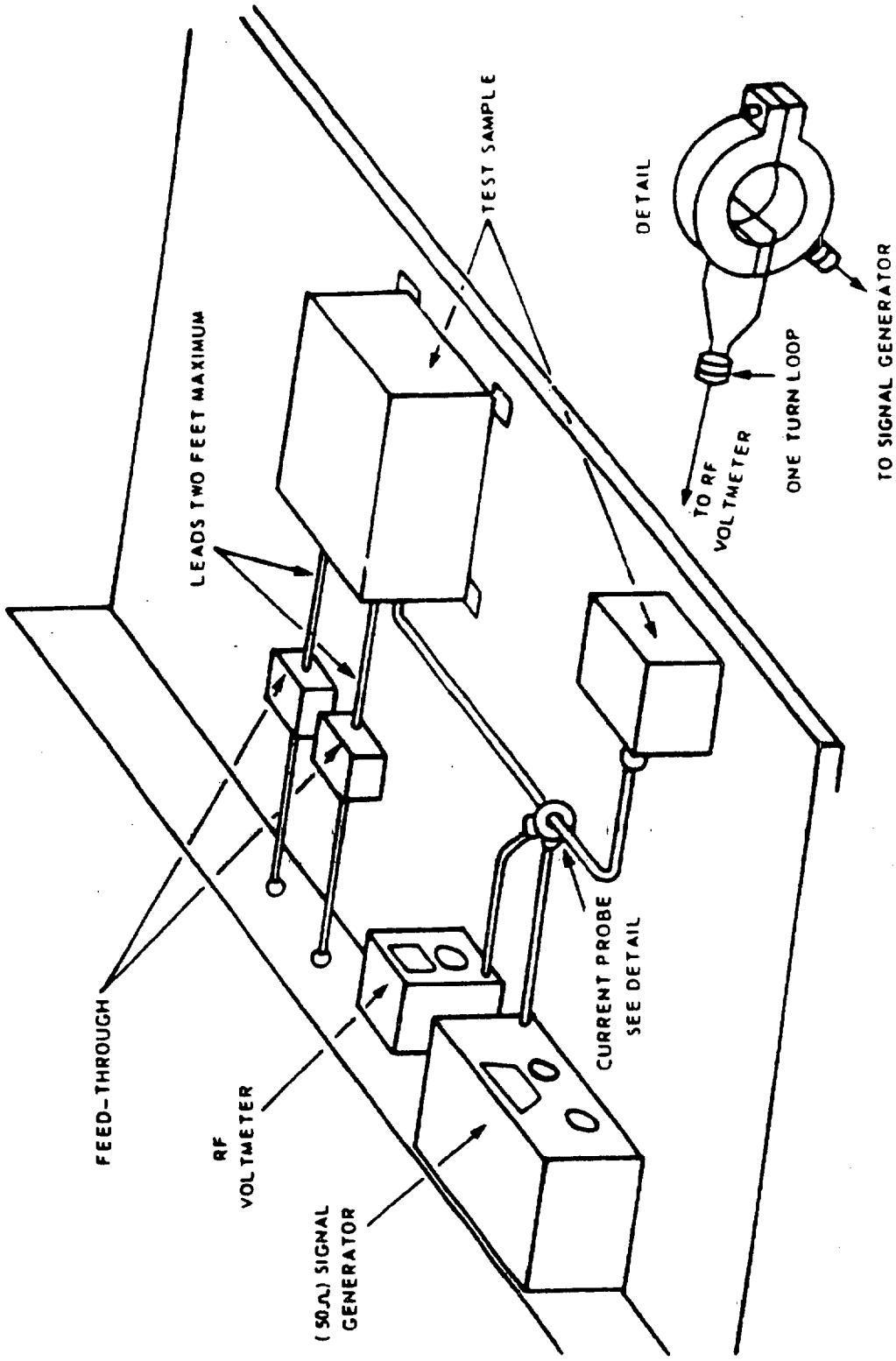


FIGURE 9. Conducted susceptibility test set up

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

c. Record and include in the test report the threshold of susceptibility data (frequency, amplitude, and description of responses) obtained in (b) above. Also include in the test report the corrective action required for each wire that does not meet the required level.

4.4.3.3.1 Wire bundles. Wire bundles may be tested in sections if the current probe will not fit around the complete wire bundles. Chassis or circuit grounds not normally routed with the interconnecting wiring shall be measured individually and shall not be included in the wire bundle under test. Power ground return leads shall not be placed in the probe.

4.4.4 Random vibration test. A random vibration test shall be conducted as specified in 4.4.4.1 through 4.4.4.4.

4.4.4.1 Test item operation. The test item shall be operated during application of random vibration so that the functional effects caused by these tests may be evaluated. The test item shall meet performance requirements, as specified, while the functional vibration levels are being applied and immediately preceding and following the application of the endurance levels.

4.4.4.2 Mounting techniques. The test item shall be hard mounted either directly to the vibration exciter or transition table, or by means of a rigid fixture capable of transmitting the vibration conditions specified herein. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of extraneous responses in the test setup. The test load shall be distributed as uniformly as possible on the vibration exciter table in order to minimize effects of unbalanced loads. The input control sensing device(s) shall be rigidly attached to the vibration table, or fixture if used, as near as possible to the attachment point(s) of the test item. Additional vibration sensors shall be located in or on the test item to determine resonant frequencies and amplification factors. Locations to be selected should include main structure, printed circuit boards, large components, and modules, where practicable. The sensor sizes and weights shall be limited so that their effect on the dynamic responses being measured is minimal.

4.4.4.3 Performance of test. The individual equipment test item shall be subjected to broadband random vibration excitation with the power spectral density envelope shown on Figure 10. The test item shall be attached to the vibration exciter according to 4.4.4.2. Vibration shall be applied sequentially along each of the three orthogonal axes of the test item. Two test levels are required, a functional level and an endurance level. For each axis, one half of the functional test shall be conducted first, then the endurance test, followed by the second half of the functional test. The equipment shall perform according to the operating requirements as specified in 4.4.4.1. The acceleration power spectral density (g^2/Hz) of applied vibration, as measured on the test fixture at mounting points of the test item, shall be as specified in 4.4.4.4. Test times shall, for each axis, be one hour each for functional and endurance levels. The instantaneous random vibration acceleration peaks may be limited to three times the rms acceleration level. The power spectral density of the test control signal shall not deviate from the specified requirements by more than +100, -30

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

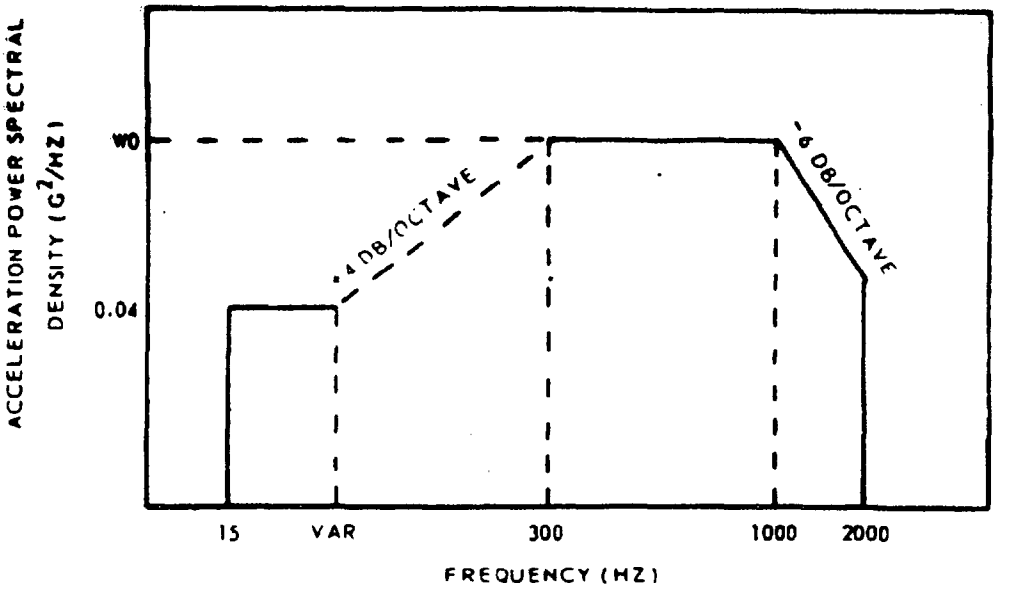


FIGURE 10. Random vibration envelope

percent (+3, -1.5 dB) below 500 Hz and +100, -50 percent (+3 dB) between 500 Hz and 2,000 Hz except that deviations as large as +300, -75 percent (+6 dB) shall be allowed over a cumulative bandwidth of 100 Hz maximum, between 500 and 2,000 Hz. Tolerance levels in terms of dB are specified as:

$$dB = 10 \text{ Log}_{10} \frac{W1}{WO}$$

Where W1 = measured acceleration power spectral density in g^2/Hz units, the term WO defines the specified level in g^2/Hz units. Confirmation of these tolerances shall be made by use of an analysis system providing statistical accuracies corresponding to a bandwidth-time constant product, BT=50 minimum. Specific analyzer characteristics shall be as specified below or equivalent, subject to the BT=50, minimum limitation.

a. On-line, contiguous filter, equalization/analysis system having a bandwidth as follows:

B=25 Hz, maximum between 20 and 200 Hz
B=50 Hz, maximum between 200 and 1,000 Hz
B=100 Hz, maximum between 1,000 and 2,000 Hz

b. Swept frequency analysis systems characterized as follows:

(1) Constant bandwidth analyzer.

(a) Filter bandwidth as follows:

B=25 Hz, maximum between 20 to 200 Hz
B=50 Hz, maximum between 200 to 1,000 Hz
B=100 Hz, maximum between 1,000 to 2,000 Hz

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:51Z
Check the source to verify that this is the current version before use.

(b) Analyzer averaging time =T= 2 RC = 1 second, where T= true averaging time constant

(c) Analysis sweep rate (linear) =

$$R = \frac{B}{4RC} \text{ or } \frac{B^2}{8} \text{ (Hz/second)}$$

(2) Constant percentage bandwidth analyzer

(a) Filter bandwidth - pfc=one-tenth of center frequency maximum (0.1 fc) where p=percentage and fc=analyzer center frequency

(b) Analyzer average time = T = $\frac{50}{pfc}$, minimum

(c) Analysis sweep rate (logarithmic) =

$$T = \frac{pfc}{4RC} \text{ or } \frac{(pfc)^2}{8}$$

(Hz/second), maximum, whichever is smaller.

Digital power spectral density analysis employing quantization techniques providing accuracies corresponding to the above approach.

4.4.4.3.1 Worst case axis. During the vibration testing, data shall be obtained to identify the worst case axis for purpose of vibration screening tests on production equipments.

4.4.4.4 Test levels. For the purpose of the random vibration test, the following test levels shall be used:

a. Functional Level-W_o = 0.05 g^2/Hz (8.5g rms minimum)

b. Endurance Level-W_o = 0.12 g^2/Hz (12.7g rms minimum)

4.4.5 Reliability qualification test. A reliability qualification test may be included in the first article tests. A maximum of six controls shall be tested in accordance with MIL-STD-781 using the reliability qualification test - mission profile in this specification. (See Figures 11 and 12.) Each control shall have successfully completed the normal burn-in and all equipment test (AET) prior to the start of RQT. The mission profile test cycle should begin at the "12 hour" point which is a cold day ground stabilization (-54°C) prior to ground operation and flight in a cold day. Random vibration should be applied with the units hard-mounted (without vibration isolators). The accept-reject criteria imposed shall be those of MIL-STD-781, Test Plan IIIC, with the upper test MTBF specified in 3.4.1, except that the test shall be truncated at 10,300 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

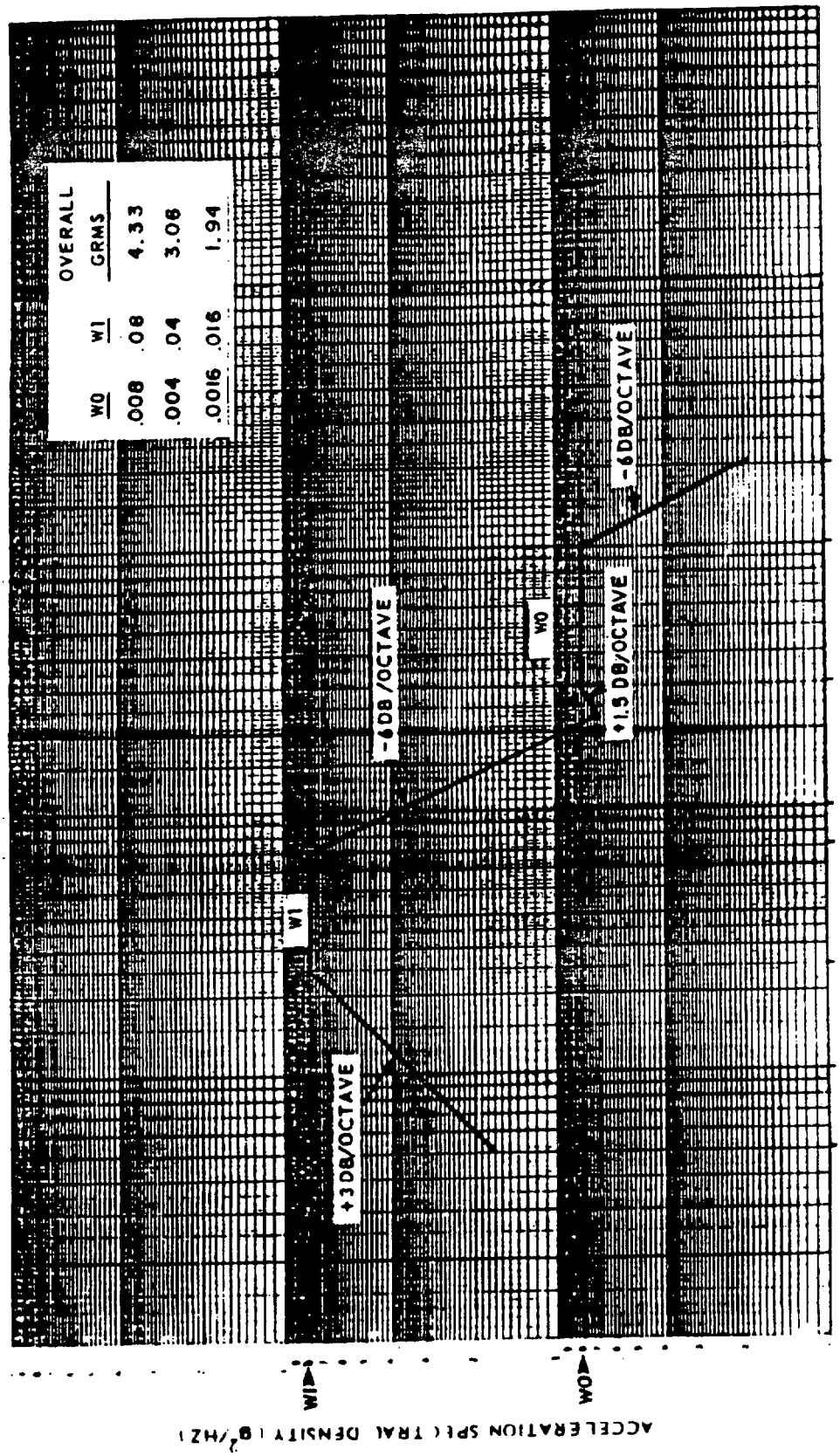


FIGURE 11. RQT vibration envelope (see 4.4.5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

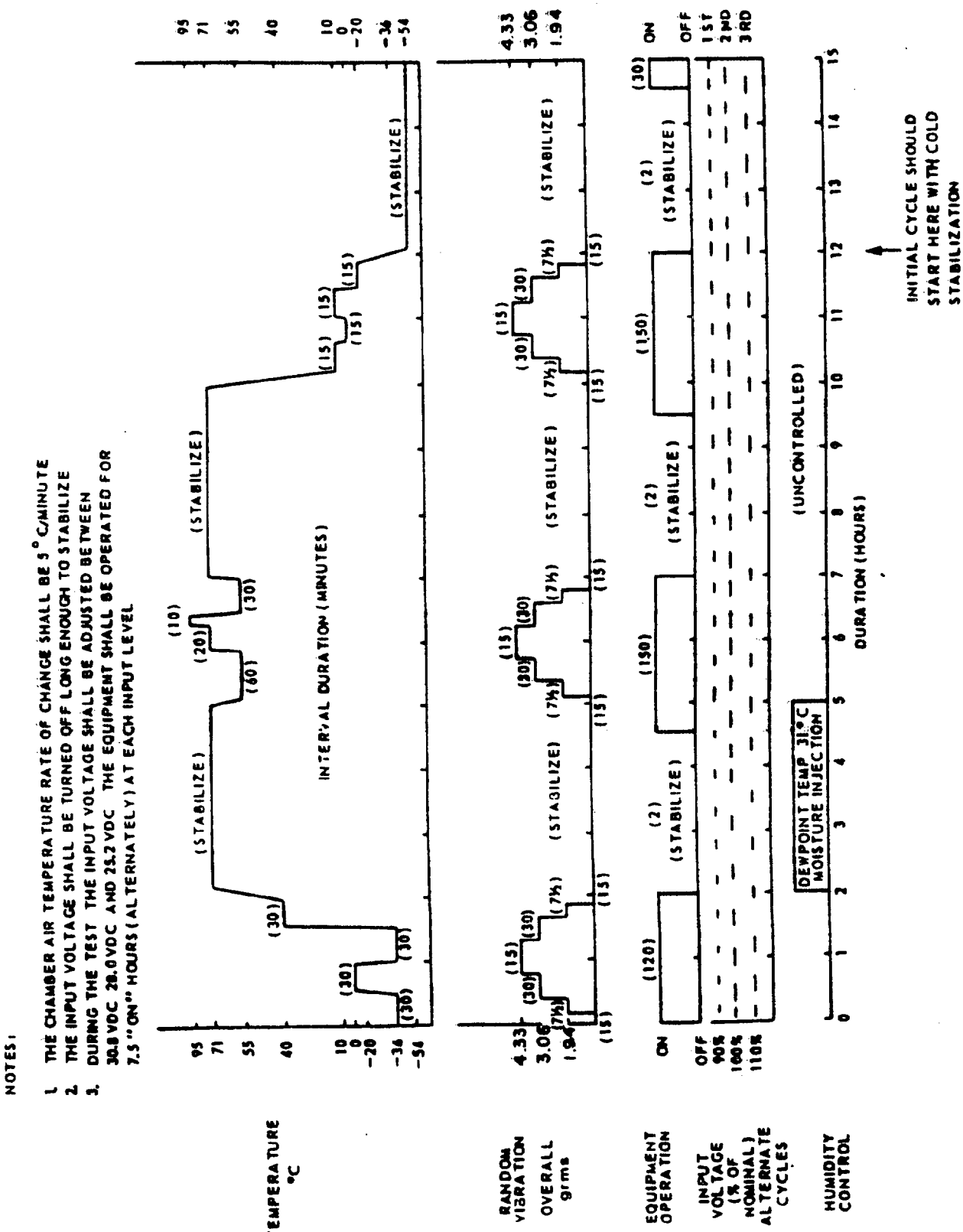


Figure 12. Temperature/power cycle (see 4.4.5).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

4.4.6 Maintainability demonstration. A maintainability demonstration for intermediate level requirements shall be conducted in accordance with MIL-STD-2084 and MIL-STD-471. Test Method 1, Test Plan B of MIL-STD-471, shall be used.

4.4.7 Production controls. Controls supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be capable of successfully passing the same tests as specified herein.

4.5 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance tests. All inspection and testing may be under the supervision of the customer representative. The contractor shall retain test data showing quantitative results for all quality conformance tests. Such tests shall be signed or stamped by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance tests shall consist of the following:

a. Individual tests (see 4.5.1)

b. Manufacturing screening (see 4.5.2)

c. Sampling tests (see 4.5.3)

d. Reliability assurance tests (see 4.5.4)

e. Special tests (see 4.5.5)

4.5.1 Individual tests. Each control submitted for acceptance shall be subjected to the individual tests. The tests shall be adequate to determine compliance with the requirements of materials, workmanship, operational adequacy and reliability. As a minimum, each control accepted shall have passed the following tests:

a. Examination of product (see 4.5.1.1)

b. Operational test (see 4.5.1.2)

4.5.1.1 Examination of product. Each control shall be examined carefully to determine that the material and workmanship requirements have been met.

4.5.1.2 Operational test. Each control shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record data to assure specified equipment operation.

4.5.2 Manufacturing screening. The contractor shall subject each WRA to manufacturing screening prior to undergoing sample tests or reliability assurance tests. Manufacturing screening shall consist of WRA random vibration and temperature cycling. An additional screening period may be used at the option of the contractor. If an additional screening period is to be used, the details thereof shall be included in the approved test procedures and shall be the same for all equipment, including spare units.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

4.5.2.1 WRA random vibration. Prior to conducting WRA temperature cycling, each WRA shall have completed a 10 consecutive minute failure-free vibration screens at 6.1 gs. Power spectral density shall cover 20-2000 Hertz. Pseudo-random vibration methods are acceptable. Vibration envelope is shown on Figure 13. The test item shall be hard mounted without vibration isolators and shall be operating throughout the test.

4.5.2.2 WRA temperature cycling. Each WRA shall undergo 25 temperature cycle between -54°C and 71°C. The rate of temperature change between extremes shall be at least 5°C per minute. When the WRA temperature has stabilized within 5°C or either temperature extreme, the WRA shall be soaked. Hot soak at +71°C shall be two hours; cold soak at -54°C duration shall be one hour. At the conclusion of each soak, the next temperature ramp shall begin. The WRA shall be energized throughout temperature cycling except during the cooldown period from +71°C to -54°C and during the 60 minute soak at -54°C. In addition, the WRA shall be turned off for one minute at points 30, 60, and 90 minutes into each hot soak period. The controls performance shall be periodically checked during each cycle and repairs shall be made to correct failures. The last five consecutive temperature cycles shall be failure free. One temperature cycle is defined as: operation at +71°C for the 120 minute hot soak (including three minutes "off"), non-operational cooldown to -54°C, 60 minute non-operational cold soak at -54°C, operational temperature increase to +71°C.

4.5.3 Sampling tests. Controls shall be subjected to sampling tests. Controls selected for sampling tests will first have passed the individual tests. Controls shall be selected for sampling tests by the government inspector in accordance with Table XV.

TABLE XV. Sampling test sample selection

Quantity of equipments offered for acceptance	Quantity to be selected for sampling test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

Sampling tests are not required when reliability assurance tests are conducted.

4.5.3.1 Scope of sampling tests. As a minimum, each control selected for sampling tests shall be subjected to the following tests:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

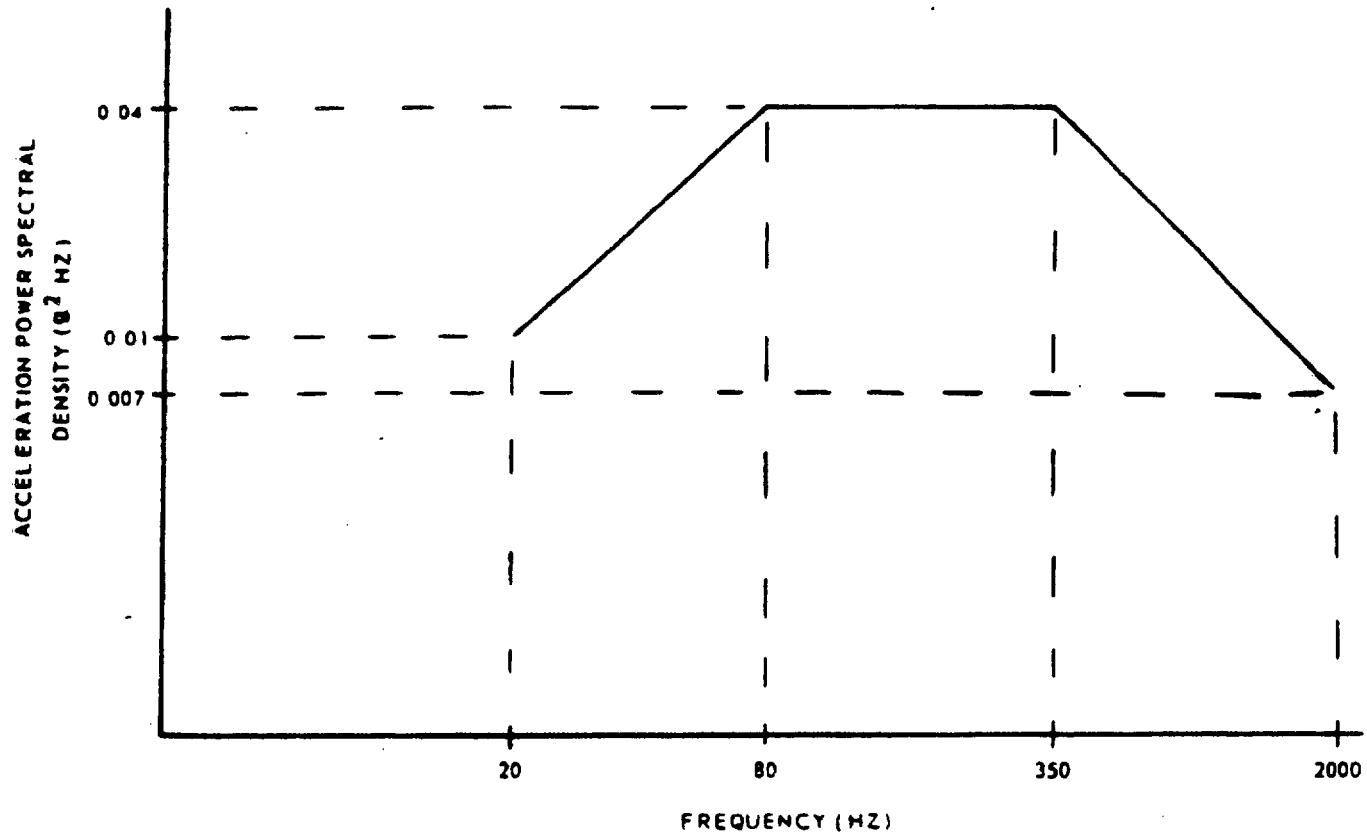


FIGURE 13. Manufacturing screening random vibration envelope

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each control tested and should be based on results of the first article, individual and special tests.

c. Reliability sample test specified in 4.5.4.2. Each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurring in future equipment taken.

4.5.4 Reliability assurance tests. A reliability assurance test shall be conducted using MIL-STD-781. Classification of failure shall be in accordance with 4.5.7.

4.5.4.1 All equipment test. Each control shall be tested as specified in MIL-STD-781 and modified herein. Temperature cycling and power on/off cycling shall be as shown on Figure 12. No vibration is required. Each control shall have successfully completed manufacturing screening as specified in 4.5.2 prior to the start of the test. Each equipment (WRA) shall be tested for 125 equipment operating hours with the last two cycles failure free. For determining MTBF compliance each equipment (WRA) shall be tested separately as opposed to testing as an equipment system. The test MTBF shall be as listed in 3.4.1.3. To determine whether the MTBF is being met at any time during the contract the operating time and the failures therein (not counting any manufacturing screening failures or operating time) shall be totaled and the results compared with the reject line of Test Plan XVIIIC of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any time the experience from the beginning of the contract is included. At any time that the current totals of test hours and test failures plotted on Test Plan XVIIIC curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of control at any time that a reject situation exists pending a review of the contractors efforts to improve the equipment, the equipment parts, the equipment workmanship, so that the entire compilation will show other than a reject decision.

4.5.4.2 Reliability sample test. When specified in the contract (see 6.2.1), a monthly sample test of equipments shall replace the reliability test of 4.5.4.1. Units shall be tested as specified in MIL-STD-781 as modified herein. Temperature cycling and power on/off cycling are required and shall be as shown on Figure 12. Each sample equipment (WRA) shall be tested for 256 hours with the last two cycles failure free. For determining MTBF compliance each designated equipment (WRA) shall be tested separately as opposed to testing as an equipment system. The MTBF for each designated equipment shall be those listed in 3.4.1.3. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures therein (not counting any burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan IIIC of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. The sample size shall be based on the production quantity of that equipment for the month as specified in Table XVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE XVI. Reliability test sample size

Production Quantity	Sample Size
2 - 5	1
6 - 25	2
26 - 50	3
51 - 100	4

4.5.4.3 Test details. Test details such as the performance criteria to be measured, special failure criteria, and preventive maintenance to be allowed during the test, shall be part of the test procedures to be prepared by the contractor prior to the beginning of the reliability assurance tests.

4.5.5 Special tests. When specified in the contract, special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the control and to assure quality control (see 6.2.1). The control selected for special tests may be selected from controls previously subjected to the sampling or reliability tests.

4.5.5.1 Special test schedule. Selection of controls for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.5.5.2 Scope of special tests. Special tests shall consist of such tests as authorized by the procuring activity. Test procedures previously approved for the first article tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure for the procuring activity acceptance prior to conducting the tests.

4.5.6 Equipment failure. Should a failure occur during either the sampling, reliability qualification, reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure and the equipment BIT performance in detecting and isolating the failure.

b. Determine if the failure is an isolated case or design defect or pattern failure.

c. Prepare proposed corrective action intended to reduce the possibility of the same component or BIT failure(s) occurring in future tests or documented justification where no corrective action is proposed.

d. Where practical, include a test in the individual test to check all controls for this requirement until assurance is obtained that the defect has been corrected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

4.5.7 Failure classification. All incidents occurring during reliability assurance tests shall be classified and reported as either relevant or non-relevant in accordance with MIL-STD-2074 and MIL-STD-781. Only those incidents classified as relevant failures shall be accountable for making an accept or reject decision.

4.5.7.1 Relevant failures. All test incidents shall be considered as relevant failures unless determined to be non-relevant in accordance with the criteria of MIL-STD-2074.

4.5.7.2 Non-relevant failures. Although non-relevant failures are not used for reliability assurance test accept or reject decisions, all test article incidents and equipment failures shall be recorded and reported.

4.6 Test procedures. The procedures used for conducting all tests required by Section 4 shall be prepared by the contractor for review and acceptance. The right is reserved by the procuring activity or the government representative to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of the specification or the contract (see 6.2.1). MIL-T-18303 shall be used as a guide for preparation of test procedures.

4.7 Testing of spares and repair parts. Equipment delivered as spare or repair parts shall undergo acceptance tests to ensure that the items meet performance and reliability requirements not less than those imposed in production.

4.7.1 Weapons replaceable assemblies. All items to be delivered as spares at the weapons replaceable assembly (WRA) or higher level shall receive performance, manufacturing screening and reliability assurance tests commensurate with that afforded the original production equipment.

4.7.2 Shop replaceable assemblies. All items to be delivered as spares at the shop replaceable assembly (SRA) or sub-SRA level shall receive functional checks and manufacturing screening under conditions and for a duration commensurate with that imposed on that assembly in the original production.

4.7.3 Repair parts. All electronic component parts to be delivered as repair parts shall receive functional manufacturing screening tests equal to those imposed for the nearest military standard part.

4.7.4 Test details. The details of acceptance tests to be performed on SRA's and WRA's supplied as spare or repair parts shall be incorporated into item specifications and drawings.

4.8 Presubmission testing. No item, part, or complete equipment shall be submitted until it has been previously tested by the manufacturer and found to comply with all applicable requirements of this specification and the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 General. All major units and parts of the controls shall be preserved, packaged, packed, and marked in accordance with MIL-E-17555 for the level of shipment specified in the contract or purchase order (see 6.2.1).

5.2 Marking. Shipments shall be marked in accordance with requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The controls specified herein are one portion of the AN/ARC-182(V) radio system. The AN/ARC-182(V) is a variable configuration of units using common modules that can be used to meet the prime and back-up UHF voice plus prime UHF/VHF/AM/FM voice and data communication requirements in any tactical aircraft. The variable configurations are intended to provide direct physical replacement of AN/ARC-159(V) radios. They are intended for production incorporation in new aircraft plus retrofit in selected aircraft.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification

b. If a first article inspection is required (see 3.2, 6.3)

c. If samples of non-standard parts and materials are not required (see 3.3.5)

d. If reliability program is not to be established by contractor (see 3.4.1)

e. Test procedure modifications or additional tests, if required (see 4.5.5 and 4.6)

f. Selection of applicable levels of packaging and packing (see 5.1)

g. Whether All Equipment Test or Reliability Sample Test is required (see 4.5.4.1, 4.5.4.2)

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL) incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph No	Data requirement title	Applicable DID No.
3.3.2	Engineering Change Proposals (ECPs) and Requests for Deviations and Waviers (Long Form)	DI-E-2037
3.3.3	Program Parts Selection List (PPSL)	DI-MISC-80072
3.4.1, 4.5.6c	Plan, Failure Data Collection Analysis and Corrective Action	UDI-T-23719
3.4.1.2	Part Approval Request	DI-MISC-80071
3.4.8	Request for Confirmation of Nomenclature	DI-CMAN-80194
	Request for Assignment of Serial Number and Serial Number Prefix Letters	DI-CMAN-80195
	Request, Assignment of Nomenclature	UDI-E-21582
	Request for Approval of Identification Plate Drawings	DI-CMAN-80196
4.4, 4.6	First Article Inspection Procedure	DI-T-4901
4.4.1, 4.4.2	First Article Inspection Report	DI-T-4902
4.4.3.1, 4.4.3.2	Electromagnetic Interference Test Plan	DI-EMCS-80201
4.4.3.3c	Electromagnetic Interference Test Report	DI-EMCS-80200
4.5.4.3	Reliability Test Procedures	DI-RELI-80251
4.5.5.2	Production/Acceptance Inspection Procedures	DI-T-4903
4.5.7.2	Failed Item Analysis Report	DI-RELI-80253
4.7.4	Drawings, Engineering and Associated Lists	DI-E-7031

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12L AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

6.3 Options. The following options are available:

a. An additional manufacturing screening period is allowable at the option of the contractor (see 4.5.2).

b. Acceptance of equipment may be stopped anytime a reject situation exists in the All Equipment Test. This is at the option of the procuring activity (see 4.5.4.2).

6.4 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Associated equipment. These controls should operate with the associated equipment listed in Table XVII which should not be supplied as part of this equipment:

TABLE XVII. Associated equipment

Item	Equipment Designation	Military Specification
Intercommunications Sets	AN/AIC-10, AN/AIC-14 AN/AIC-18, AN/AIC-25 C-6533/ARC	MIL-I-22353 MIL-I-27543 MIL-C-55653
Microphones, Boom Mounted	M-96A/A	MIL-M-22179
Amplifiers, Audio Frequency	AM-3597B/A	MIL-A-23595
Antenna	AS-3191/ARC-182(V) AS-3238/ARC-182(V) AT-256A/ARC	MIL-A-85670(AS) AS-4580(AV) MIL-A-5815
Switch	SA-521A/A with Selector Antenna C-2193B/A	MIL-S-25879
Direction Finder Group	AN/ARC-25, AN/ARC-48 AN/ARA-50, AN/ARD-6987	MIL-D-7030 MIL-D-38402
Speech Security Equipment	TSEC/KY-28 TSEC/KY-58	CSEEB-13 CSEEB-32/CSEEB-14
VHF Homing Indicator	ID-1351/A	SCS-405

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE XVII. Associated equipment - Continued

Item	Equipment Designation	Military Specification
Radio Set, Survival	AN/PRC-90	MIL-R-81493
Indicator, Frequency/Channel	ID-2121A/ARC	MIL-I-85671(AS)
	ID-2229A/ARC	MIL-I-85671(AS)
	ID-2303/ARC	MIL-I-85671(AS)
Receiver-Transmitter, Radio	RT-1324A/ARC	MIL-R-85665
	RT-1327A/ARC	MIL-R-85665
	RT-1407/ARC	MIL-R-85665
Control, Radio Set	C-11628/ARC	MIL-C-84674
	C-10777A/ARC	MIL-C-84674
	C-11132/ARC	MIL-C-84674
Switching Unit	SA-2498/ARC	MIL-S-85673
Receiver-Transmitter	RT-1250A/ARC	MIL-R-85664

6.6 Subject term (key word) listing.

AN/ARC-182(v) Radio Set
Blue Filtered White Panel Lighting
Built-In-Test (BIT)
Control
Frequency Select
Mode Select
Non-anti-jam
Panel Mounted
Red Panel Lighting
Remote Control
Ultra High Frequency (UHF)
Very High Frequency (VHF)
White Panel Lighting

Preparing Activity:
Navy AS
(Project 5821N-193)

U.S. GOVERNMENT PRINTING OFFICE: 1987 - 704-036/74286

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.